ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Sonar Operator (SONAR Op) (00324)

Sonar Control Supervisor – RQ-PO2

NQUAL - ALBD

MITE ID - 121983

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance are to be approved by CO NTDC(A).

Revision	Approval Date	Authority
1.0	08 May 2020	NPTG HQ
1.1	17 November 2020	CO NTDC(A)

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FOREWORD

1. This training is based on the specification found at :

   a. A-PD-055-002/PP-002, SONAR OP, 00324 Occupation Specifications, dated 20 Mar 18.

2. A trainee will achieve the qualification NQUAL ALBD when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve MITE 121983 when standards listed in the associated Performance Objectives have been met.

4. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A) / NFS (P)	All	121983	ALBD

CHAPTER 1: TRAINING MANAGEMENT DETAILS

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AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the RQ-PO2 SONAR Op NQUAL ALBD. This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	471
Enabling Checks (ECs)	58
Performance Checks (PCs)	78
Actual Training Time (ATT) Sub-total	**75.9 trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	19
Training Administrative Time (TAT)	19
Training Travel Time (TTT)	NIL
Physical Fitness (PT)	51
Sub-total	**11.1 trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	**87** **trg days**

REQUIRED COURSE STAFFING

3. Identify key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
Snr Instructor - UWWD	PO1/ Lt(N)	ALBV	1	002.01A-C & E, EC 002.01, 003.03 A-D & F-I
RQ-PO2 Instructor	PO2	ALBD	1	ALL

Guest Lecturers				
Position	**Rank**	**Qualifications**	**Qty**	**PO / EO**
TDO/NTDC staff	PO1<	Any	1	001.01A & 001.02D
CCFL/P or Sea Trg	PO1<	Any	1	001.01D
ADAC	MS <	ALBC	1	001.02C
NCIOP	MS<	ALBF	1	003.01A & C
CANSUBFOR	Any	Any	1	003.01E
IUSS member	MS <	Any	1	003.03J

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 12;

   b. Minimum – 6; and

   c. Optimum – 8.

PREREQUISITES

5. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

   a. RQ-MS SONAR Op (ALBC);

   b. 12 months of operational experience with 60 days at sea required after being awarded the RQ-SONAROP-MS (ALBC) qualification. (less than 12 months experience requires Waiver Authority IAW NPI 13 SONAR OP 00324)

COURSE REPORTING

6. Upon completion of the course, students shall receive a paper record of completion by means of form CF 377 Course Report and Form CF 289 (or locally produced equivalent) Certificate of Military Achievement IAW DAOD 31-9 and NPTORD 4500-51.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

7. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

8. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

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PO 001 – Perform the Duties of a Training Petty Officer

1. **Performance**: Perform the Duties of a Training Petty Officer

2. **Conditions**:

   a. Given:

      (1) Subordinates;
      (2) References
      (3) DWAN Computer equipped with:
         (a) Monitor mass;
         (b) Microsoft Office; and
         (c) Watch and station bill template.

   b. Denied: NIL

   c. Environment:

      (1) Any (at sea/alongside/shore facility & day/night)

3. **Standard**: The SCS shall perform the duties of a training Petty Officer by:

   a. Planning/Designing training exercises (G0010) by:

      (1) Identifying individual/team/ship training requirements;
      (2) Identifying section resources to include:
         (a) Equipment
         (b) Personnel; and
         (c) Expendables;
      (3) Coordinating with outside units

   b. Conducting unit OJT, SDP, PIP and continuation/proficiency training (G0085, G0080) by:

      (1) Conducting individual training events (G0050) by:
         (a) Assisting in individual training events (G0045);
         (b) Assessing individual training performance (G0115); and
         (c) Correcting individual training deficiencies (G0055);
      (2) Conducting team training events (G0065) by:
         (a) Assisting in team training events (G0060); and
         (b) Correcting team training deficiencies (G0070).
      (3) Compiling/distributing OJT, SDP and PIP reports (G0040); and

(4) Providing recommendations to PRB/TRBs (G0100);

4. **Training Limitations**: Nil.

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EO 001.01 - Plan/Design training exercises

001.01.01 – Occupational Progression

Conditions		Sequence
Unclassified Classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	Guest Lecture: NTDC or TDO

Learning Objectives	Method
1) Military Employment Structure 2) Occupational Structure 3) Trade Progression 4) Job Based Specifications 5) Master Task List	Guided lesson (web based) Individual Reading (JBS) Three Star

Activity Steps	
	1) Guided web based lesson, demonstrating the DPGR webpage and how to find the MES and Occupational Specifications. Instructor to open the JBS, discuss what is contained in the document and specifically talk about the MTL (open it and show what is taught at what level based on the MTL). (20 min) 2) Students are then provided with a copy of the SONAR Op Job Based Specification to read (online or paper copy). (15 min) 3) Students are then asked to create a list of their TOP THREE (three star) points that they took away from the JBS. (3 min) 4) Once students have created their lists, they are then asked to share their answers with the class (1 per student, no repeats or if time permits all 3 from each student). (7 min) 5) Instructor to conduct short close out discussion (5 min) a. How does this information support you in your role as the Trg PO?

Resources	References
1) DWAN computer 2) projector 3) JBS handout	1) DPGR website 2) Occupational Specifications SONAR Operator MOSID 00324

Notes to Instructor
Activity details found in AIT manual If students do not have access to DWAN, ensure you have a paper based copy of JBS for each student.

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EO 001.01 - Plan/Design training exercises

001.01.02 – RQ-S2 On the Job Progress Report

Conditions		Sequence
Unclassified Classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) OJPR document layout 2) QSP Annex OJPR 3) Training package review schedule	Guided lesson (web based) Student presentation

Activity Steps	
	1) Instructor to demonstrate to students where to find the QSP online (NPTG website/TDC(A)) and discuss the training authority for granting the qualification. Identify the training review schedule and requirements for extensions or lost OJPRs (15 min). 2) Students are provided a copy of the SONAR Op RQ-S2 OJPR and assigned a PO to review. (5 min) 3) Each student to then present their PO with a basic explanation of what is involved in a trainee attaining that Performance Objective. Students may provide examples based on their experience related to OJPR training (25 min). 4) Instructor to lead confirmation to include (5 min): a. What do you do if one of your trainees loses their OJPR? b. How often must the training packages be reviewed? c. What are the actions conducted when a trainee completes their OJPR?

Resources	References
1) DWAN computer with projector 2) SONAR Op RQ-S2 OJPR	1) SONAR Op RQ-S2 QSP

Notes to Instructor
SOP for qualification being granted not yet approved (NTDC(A)), instructor to inquire and use that as a reference once available.

EO 001.01 - Plan/Design training exercises

001.01.03 – Combat Readiness Requirements (CRRs)

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Using CFCD 102 2) UWW applicable CRRs 3) Training requirements (personnel, equipment, facilities, time, resources) 4) Validity period/maintenance 5) Identifying authority for CRR completion	EDI Student presentations

Activity Steps

1) Instructor to introduce the publication CFCD 102. (10 min)
    a. Explain use of publication (tracking of unit readiness)
    b. Identify that Sea Training Group is the authority/maintainer of CFCD 102
    c. Demonstrate how to determine the applicable chapters
    d. Indicate how to disseminate CRR status (inform IMD)
2) Students are then paired up and assigned (minimum) 2 UWW CRRs per group and asked to identify findings to include: (20 min)
    a. CRR title
    b. Purpose
    c. Readiness level
    d. Validity period
    e. Training requirements (resources, personnel, etc.)
    f. Standard (assessment criteria)
    g. Assessment Authority
3) Each group to present their findings to the class (15 min)
4) Instructor to reaffirm the purpose of CRRs and how this will be applicable in their role as the Training PO. (5 min)

Resources	References
1) DWAN computer x 12 2) Projector 3) CFCD 102 (web based)	1) CFCD 102 2) Readiness Training Guide

Notes to Instructor

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EO 001.01 - Plan/Design training exercises

001.01.04 – Ship/Fleet Training

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	Guest lecture from CCFL/P or Sea Training

Learning Objectives	Method
1) Readiness level training requirements 2) Ships Tiered Readiness Program a. TRI 1 & 2 b. OTT 1 & 2 c. SRIs 3) Fleet Training requirements (CREST)	Lecture

Activity Steps	
	1) Guest lecturer to present on the LOs identified (40 min) a. Ships Tiered Readiness Program b. Fleet Training Requirements 2) Show and explain example for Fleet Training messages (10 min)

Resources	References
1) DWAN computer 2) projector 3) example messages for fleet training	1) NAVORD 11000-1 2) NAVORD 3137-2 3) SSOs 4) CFCD 102 5) CFCD 129 6) Readiness Training Guide

Notes to Instructor
Liaise in advance with guest lecture to ensure their presentation meets LO requirements.

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EO 001.01 - Plan/Design training exercises

001.01.05 – Section resources - Equipment and expendables

Conditions		Sequence
Unclassified Classroom		
Time	**Classification**	**Instructor**
100 mins	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Identify section training equipment 2) Familiarize with Basic Ammunition Allocation List requirements 3) Identify requirement for expendable reports	T-Chart Lecture

Activity Steps

1) Instructor to introduce lesson, indicating that you will be using the students experiences to identify the section training resources/equipment IOT demonstrate what training can be done with each asset they have. (5 min)
2) Instructor will group students into 3 groups to conduct a T-Chart on UWW training equipment. Students are to create a list of section training equipment and identify what training is conducted using that equipment. (30 min)
   Anticipated Responses:

Training Equipment	Training conducted
ETS	HMS Operator Trg
CANTASS Training Tape	CANTASS recognition/tracking
PTP/EMATT	Programming a target
Osprey	Passive Acoustic Analysis training
NITES2R	Acoustic Range Prediction Trg
Training Shapes: ADC NAEB Sonobuoy ESUS Bathy	Equipment familiarization/CRRs/OJPR
SETT/SAFSIM	Basic/Advanced ASW Trg/CMS trg
UWT	Communications training/ OJPR
MK8F/MK21	OJPR

3) Groups to present equipment and training uses to class, skipping over any duplicate info (30 Mins)
4) Instructor to present BAAL to students and explain the expendable requirements based on readiness level of the ship (10 Mins).

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	5) Instructor will then show an example of expendable reporting methods/requirements and explain how to create them/track the info (15 Mins) 6) Instructor to summarize lesson objectives (10 Mins).

Resources	References
1) Flip Charts/White board 2) Assorted Markers 3) Copy of BAAL 4) Expenditure report	1) CFCD 129 Annex A (BAAL) 2) CFCD 102

Notes to Instructor
Instructor to, based on class size, pair or group students no more than three groups Class may come up with more than what is included in the anticipated responses section, this is for instructor guidance. Capitalize on class experience.

EO 001.01 - Plan/Design training exercises

001.01.06 – Section resources - Personnel

Conditions		Sequence
Unclassified Classroom		
Time	**Classification**	**Instructor**
200 Mins	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Identify section REMAR 2) Identify UWW Section crewing requirements 3) Identify Ship Special Parties 4) Develop Watch and Station Bill a. Full REMAR b. Reduced crewing	Interactive lecture Jigsaw EDI Collaborative Problem solving

Activity Steps	
	1) Instructor to provide students with REMAR and positions, explaining the significance of this WRT crewing (5 Mins). 2) Students are then broken into 3 equal groups to conduct a jigsaw activity. Each group will be assigned a specific reading list, where they must become the experts on the readings. (35 min) Topics/Readings: 1. SSOs Chapter 2 2. SSOs Chapter 4 3. CFCD 129 Chapter 4, articles 404, 406, 411, Annex A & B Once groups have become "experts" in their reading section, students will be numbered 1, 2 or 3 (only 1, 2 for classes of 6 or less). Groups will reform (all 1s to one area, all 2s to another, etc.) and students are to teach each other the content from their readings section, covering all key points. (30 min) 3) Instructor to conduct and EDI lesson on watch and station bill. Demonstrating the use Sea Trg Group webpage where to find the template and discuss the specific positions that should be manned. Be sure to demonstrate how to complete the special parties tab (only use one tab to allow the auto fill function to work).(15 min) 4) Students are then to break into pairs (or threes) and create a W/S bill (60 min) using a provided scenario (crewing) with specific considerations for: a. 1st degree Crewing requirements b. 2nd degree Crewing requirements c. 4th Degree Crewing requirements d. Special Parties 5) Once students have created their W/S bill, instructor to move to each group and discuss their plan. Provide ideas to solve any issues they may have encountered. (5 min)

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	6) Students are then told to modify their current plan with a reduction in crewing of 3 personnel (additional scenario). (30 min) 7) Once each group has completed their modifications, the instructor is to pull up one groups W/S bill on the projector and have the class discuss the plan. (Identify gaps, good considerations, etc.). (20 min)

Resources	References
1) DWAN terminals (12) 2) Projector 3) Watch and Station Bill template 4) REMAR lecture 5) Watch and Station Bill scenario (full crewing) 6) Scenario addition (reduction in crewing)	1) SSOs 2) CFCD 129 3) Sea Trg Group website

Notes to Instructor

For more information on how to conduct a JIGSAW activity, see Alternate Instructional Techniques manual.
Building a W/S bill is collaborative problem solving, as there is not definitive “RIGHT ANSWER”, identify areas to consider in utilizing personnel optimally.

EO 001.01 - Plan/Design training exercises

001.01.07 – Scheduling Training Events with outside units

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Scheduling trainer facilities (Campus Atlantic/Campus Pacific) 2) Liaising/scheduling with ADAC 3) Liaising/scheduling with METOC 4) Identify support centres capabilities (CFMWC, DRDC, BIO, TRINITY Intelligence, etc.)	Think/Pair/Share Collaborative T-Chart Directed lesson (scheduling)

Activity Steps

1) Instructor to ask students to individually create a list of the facilities/units they believe they would need to contact to coordinate/schedule training in their role as a Training PO. (4 min)
2) Once each student has created a list individually, they are directed to pair up and discuss their list with a peer. (3 min)
3) Students are then asked to create a class list (share their list). This list will be put up on the white board, for a class T-chart activity. (3 min)
4) Once the class list is created, instructor to lead the class in identifying what support these units provide. (20 min) Anticipated responses may be:

Units	Training Supports Provided
ADAC METOC CFMWC DRDC BIO (route survey) TRINITY/ATHENA Campus Atlantic/Pacific	Acoustic Analysis Trg/Post analysis feedback Acoustic Range prediction trg/bathy feedback Allied/Canadian Doctrine, lessons learned Future products training Bottom mapping, MCM, UUVs Training Intelligence Trainers, classrooms, facilities

5) Instructor to conduct a directed lesson on booking the various units (POC, lead time requirements, booking process, special considerations). Walk through Campus booking tools and calendars. (20 min)

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Resources	References
1) DWAN computer (instructor) 2) projector 3) contact list handout 4) whiteboard with markers	1) Campus Websites 2) Unit websites

Notes to Instructor

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EO 001.01 - Plan/Design training exercises

001.01.08 – Developing a Training Calendar

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
100 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Building a training calendar	EDI essence

Activity Steps	
	1) Instructor to demonstrate how to complete a 1 month training schedule/calendar for the section. Identifying prioritization of CRRs, OJPR, ships schedule, units to be booked, holidays/weekends, individual/team training, member specific trg requirements). (20 min) 2) Students are then paired up and provided a scenario outlining training requirements and must create a 3 month training plan for the section. (70 min) 3) Instructor to float between the groups, assist when there are difficulties and discuss any challenges (provide feedback). 4) Students are then asked to conduct an essence exercise in their pairs. a. Students are asked to write down in 20 words or less their priorities/considerations when creating a training calendar. (5 min) b. Next, students are asked to chop that sentence, list in half to only 10 words to capture the salient/important points about creating a training calendar. (5 min)

Resources	References
1) DWAN computers (12) 2) projector 3) training calendar template 4) scenario (expired CRRs, OJPR, ships schedule)	1) CFCD 102 2) RQ S2 QSP OJPR 3) SSOs 4) CFCD 129

Notes to Instructor
For more details on how to conduct activities, refer to AIT manual.

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EC 001.01 - Plan/Design training exercises

Scope	Conditions	Sequence
EO 001.01	Unclassified Classroom	O/C EO 001.01
Time	**Classification**	**Instructor**
100 min	Unclassified	RQ-PO2 Instructor

Scenario	Ship is ramping up for deployment and RQ-PO2 student is acting as the training PO for the section. They must create a training plan for the section taking into consideration the ships schedule and training needs of the members and team in preparation for deployment.
Assessment Steps	Input: Students provided with: - Ships 3 month schedule - RQ-S2 OJPR partially completed - Current CRR tracking for unit - List of section personnel (including qualifications) - NETP training plan/schedule Performance: Students are provided 100 min to create a 3 month training calendar scheduling for individual and team training needs indicating prioritization for CRRs. Output: Creation of a 3 month training calendar Feedback: Feedback to be provided to the student using EC 001.01 checklist in written form within 48hrs.

Resources	References
1) DWAN workstation with: a) Microsoft Office b) calendar template 2) Scenario (ships schedule, CRR requirements, OJPR)	1) CFCD 102 2) SSOs; 3) RQ-S2 OJPR; 4) NAVORD 4500-15

Notes to Instructor
If DWAN workstations not accessible instructor to provide hard copy calendars to be filled out by hand by students.

Limitations	Not reflective of realistic last minute ships schedule changes in training environment.
Impact of Failure	IAW NPTORD 4500-54

Item	EC 001.01 - PLAN/DESIGN TRAINING EXERCISES	SAT	UNSAT	Remarks
1. Plan/design Training Calendar				
All expired CRRs are scheduled		☐	☐	
Alongside CRRs scheduled during a/s period		☐	☐	
At sea CRRs scheduled during at sea time		☐	☐	
CRR opportunities are maximized		☐	☐	
Unscheduled/Scheduled CRRs are substantiated		☐	☐	
Individual training scheduled for subordinates		☐	☐	
OJPR progress review scheduled		☐	☐	
External units to book are identified		☐	☐	
Resources for personnel are identified		☐	☐	
Resources for equipment are identified		☐	☐	
Resources for expendables are identified		☐	☐	
Schedule conflict resolution is adhered to		☐	☐	

Score: / 12

The trainee must achieve 9/12 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 001.02 - Conducting unit OJT, SDP, PIP and continuation/proficiency training

001.02.01 – Utilize Monitor Mass Training Tracker

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Assign training package to member 2) Update completed performance objectives in MM training tracker 3) Generating training tracker reports 4) Identify process for granting qualification	EDI

Activity Steps	
	1) Instructor to demonstrate how to assign training package to a member, how to updated completed checklist items for performance objective and how to generate a training report within Monitor Mass. (10 min) 2) Instructor to then walk the class through the same process and students are to follow along with the instructor. (10 min) 3) Students are then provided a scenario with OJPR completion dates, students are to complete the process of assigning training package, updating training and generating a report individually. (30 min)

Resources	References
1) DWAN computer (13) a. Monitor mass training/beta 2) Projector 3) Scenario	1) Monitor Mass user guide 2) RQ- S2 QSP OJPR

Notes to Instructor
Instructor to move to each student to visually confirm report is created (no requirement to print).

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EO 001.02 - Conducting unit OJT, SDP, PIP and continuation/proficiency training

001.02.02 – OJPR Evaluation Standards

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
150 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Identify Performance Objective Standards	Modified Jigsaw
2) Utilize OJPR checklists	Lecture
3) evaluate performance based on OJPR checklist	Practical Observation/Evaluation
4) providing feedback on OJPR performance	Fishbowl

Activity Steps

1) Instructor to provide students with a copy of the RQ-S2 OJPR (paper or web based). Students are broken into pairs/groups of 3 and assigned 1-2 Performance Objectives from the RQ-S2 OJPR. They must review the key components of that PO for presentation to the class. (30 min) Items to include:
    a. Performance Statement/Standard
    b. Resources Required
    c. Timings expected/required
    d. Task details
    e. Number of times/hours task must be observed
    f. Explain general outline of checklist
    g. Safety considerations (if applicable)
2) Groups to present their PO items to the class (20 min).
3) Instructor will provide some instruction on utilizing checklists, evaluating performance and how to provide positive/negative feedback. (15 min)
4) Students are then provided a copy of a checklist and a written scenario of a trainee conducting a training task. They are to evaluate the performance and draft some feedback they may give to the trainee. (20 min)
5) Instructor to ask class to share some of their feedback points and discuss (5 min).
6) Students are then provided another copy of a checklist and conduct an exercise to evaluate performance and provide feedback. (30 min) Activity details may be:
    a. Watch video (show video twice)
    b. Observe acted scenario (demonstrate twice)
    c. Use a written scenario
7) Instructor to then select two students to conduct a fishbowl. (10 min)One student will act as the supervisor and one as the trainee who completed the OJPR task. Supervisor to indicate
    a. if the student was successful in attaining the PO signature,

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	b. feedback on performance/justification and c. recommendations for improvement to performance. 8) Instructor to prompt class to provide “what might you do differently?” (20 min)

Resources	References
1) DWAN computer (instructor) 2) Projector 3) speakers 4) RQ-S2 OJPR 5) Written scenario(s) 6) OJPR checklists aligned with scenario 7) Video/script for performance	1) RQ-S2 OJPR

Notes to Instructor
First run of evaluating performance is done using a written scenario Second opportunity may be a video/acted scenario based on availability of resources (ideal), however a second written scenario may be used. For more instruction on how to conduct a Fishbowl activity, see AIT manual.

EO 001.02 - Conducting unit OJT, SDP, PIP and continuation/proficiency training

001.02.03 – Monthly Acoustic Analysis Training Packs

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	Guest Lecturer ADAC or RQ-PO2 Instructor

Learning Objectives	Method
1) Understand the standards of grading 2) Reporting monthly acoustic training	Interactive lecture

Activity Steps	
	1) Instructor to present a lecture (PowerPoint) based on LOs, encourage class discussion. (50 min)

Resources	References
1) DWAN computer 2) projector 3) monthly training pack lecture 4) ARS	1) ADAC monthly acoustic analysis training pack (to be provided by ADAC) 2) CFCD 102

Notes to Instructor

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EO 001.02 - Conducting unit OJT, SDP, PIP and continuation/proficiency training

001.02.04- PRB/TRB process

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	Guest lecturer TDO or RQ-PO2 Instructor

Learning Objectives	Method
1) PRB/TRB process 2) Training administration requirements	Individual reading Card sort Interactive lecture

Activity Steps	
	1) Students are provided the link to the NPTORD 4500-54 to read (10 min) 2) Students are broken into 2 groups to conduct a card sort. Students are to put the cards in order of the TRB process. (10 min) 3) Upon completion, the instructor will verify the answers and provide a second set of cards of OPI/responsible member for each step of the TRB. (10 min) 4) Instructor will verify the class responses and provide feedback. 5) Instructor to provide an example of a Jeopardy alert and discuss the Campus Atlantic Standing Order 4036 PRB. (10 min) 6) Students are provided a scenario with a divisional note on a trainee, and must complete a jeopardy alert for the trainee. (5 min) 7) Class to discuss their responses and identify the key information that must be captured in the training administration documents. (5 min)

Resources	References
1) DWAN computer 2) Projector 3) NPTORD 4500-54 4) Cards for TRB process 5) Cards for OPI 6) Card sort DS solution 7) Jeopardy alert document 8) Written scenario	1) NPTORD 4500-54 2) Campus Atlantic Standing Order 4036

Notes to Instructor

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EO 001.02 - Conducting unit OJT, SDP, PIP and continuation/proficiency training

001.02.05- Team Training

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
100 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Understand collective team/ship training events a. OTTs b. TRIs c. SRIs 2) SCS role in team training events 3) Threat matrix preparations 4) Recognize team deficiencies and Advice UWWD on solutions 5) Coaching subordinates in team training environment	Student self-reading Floating flip chart Interactive lecture Guided Discussion

Activity Steps	
	1) Students are provided access to NAVORD 3137-2 Annex F and CFCD 129 Chapter 4 article 411to read individually. (40 min) 2) Instructor to prepare 3 flip charts with the titles: a. OTT 1/2 considerations/requirements b. TRI 1/2 considerations/requirements c. SRI considerations/requirements 3) Students are then broken into 3 groups, assigned one flip char to start at and provided 5 min to write down what they believe are the considerations/requirements of the ASW team in preparing for these training events. After 5 min, teams are to rotate to the next chart and provided 5 min to add to the list. One more rotation, students provided 5 min to add to the list. (15 min) 4) Once all groups have had a chance to add to each chart one student will stay at a chart and present the list. Instructor to provide feedback and encourage discussion based on class experience. (10 min) 5) Instructor will conduct an interactive lecture explaining the threat matrix preparations, where to obtain the information and demonstrate a template. (10 min) 6) Instructor to lead a guided discussion regarding team training deficiencies and subordinate coaching. (25 min) Suggested instructor questions/injects a. Sea Trg has indicated a deficiency in ship adhering to the Naval Signature Management policy and noise rounds are not being conducted. What are your actions/recommendations?

	b. Ship is conducting an EMATT exercise, SPS operator fails to setup sensor in the correct configuration, which has a cascading negative effect on the CASEX serial. What are your actions/recommendations? c. Your ships receives a letter from ADAC indicating you are delinquent in your monthly training packs. What are your actions/recommendations?

Resources	References
1) DWAN workstation (13) 2) projector 3) flip charts & markers 4) threat matrix template 5) MARPAC N37-3-1	1) NAVORD 3137-2 2) NAVORD 3395-1 3) CFCD 129 4) CFCD 109

Notes to Instructor
For more information on how to conduct a floating flip chart exercise, see AIT manual

EC 001.02 - Conducting unit OJT, SDP, PIP and continuation/proficiency training

Scope	Conditions	Sequence
EO 001.02	Unclassified Classroom	O/C EO 001.02
Time	**Classification**	**Instructor**
50 min	UNCLAS	RQ-PO2 Instructor

Scenario	As the training PO for the section you are to track the individual training of your two subordinates completing OJPRs. You track the sailors in Monitor Mass and must make a recommendation on one sailors TRB based on a lack of progress.
Assessment Steps	Input: Students provided with: - 2 x RQ-S2 OJPR partially completed - Jeopardy alert template - divisional notes on both sailors Performance: Students are to input the OJPR progress of the two sailors into Monitor Mass and then complete the preparatory admin for a TRB (jeopardy alert/divisional notes) Output: Monitor Mass training tracker x 2 Jeopardy alert/Divisional notes Feedback: Feedback to be provided to the student using EC 001.02 checklist in written form within 48hrs.

Resources	References
1) DWAN workstation a) Monitor Mass b) Microsoft Office 2) Jeopardy alert template 3) scenario (OJPR, divisional notes) 4) Monitor Mass training accounts	1) RQ-S2 QSP (OJPR) 2) NPTORD 4500 -54 3) NPTORD 4500-55 4) Monitor Mass users guide 5) Campus Atlantic Standing Order 4036 PRB

Notes to Instructor

Limitations	Input of OJPR progression should be done throughout the completion of the training, inputting it all at once is not truly reflective of timings of this task.
Impact of Failure	IAW NPTORD 4500-54

Item	EC 001.02 - CONDUCTING UNIT OJT, SDP, PIP AND CONTINUATION/PROFICIENCY TRAINING	SAT	UNSAT	Remarks
1. Monitor Mass Tracker Input				
Profile created for each subordinate		☐	☐	
OJPR tracking IAW scenario		☐	☐	
OJPR progression is finalized		☐	☐	
Generate Training Tracker Progress Report		☐	☐	
2. TRB report				
Tombstone data accurate		☐	☐	
Shortcomings identified		☐	☐	
Corrective actions identified		☐	☐	
Impact of failure to meet expectations outlined		☐	☐	

Score: / 8

The trainee must achieve 6/8 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PC 001 – Perform the Duties of a Training Petty Officer

Scope	Conditions	Sequence
PO 001	Unclassified classroom	O/C of PO 001
Time	**Classification**	**Instructor**
100 min total	Unclassified	RQ-PO2 Instructor

Scenario	Ship is ramping up for deployment and RQ-PO2 student is acting as the training PO for the section. They must create a training plan for the section taking into consideration the ships schedule and training needs of the members and team in preparation for deployment. Upon submission of the schedule, students will observe a Jr operator conducting an OJPR task where the RQ-PO2 student will be required to assess and provide feedback to the Jr operator.
Assessment Steps	Input: Students provided with: 1) Ships 3 month schedule - RQ-S2 OJPR partially completed - Current CRR tracking for unit - List of section personnel (including qualifications) - NETP training plan/schedule 2) Video/acting of Jr operator conducting OJPR task - IGOSS form - Bathy message Performance: Students are provided 50 min to create a 3 month training calendar scheduling for individual and team training needs indicating prioritization for CRRs. Students will then observe a Jr Operator (video or acted scenario) conducting an OJPR task. They will take notes and provide feedback to the Jr operator on their performance. (50 min). Output: - creation of a 3 month training calendar - Feedback notes on performance of Jr Operator Feedback: Feedback to be provided to the student using PC 001 checklist in written form within 48hrs.

Resources	References
1) DWAN workstation with: a) Microsoft Office b) calendar template 2) Scenario (ships schedule, CRR requirements, IGOSS form, bathy message etc.) 3) video/scenario for acting 4) OJPR checklist 5) feedback form 6) Instructor computer with: a) projector b) speakers	1) CFCD 102 2) SSOs; 3) RQ-S2 OJPR; 4) NAVROD 4500-0 5) NAVORD 4500-15 6) NAVORD 7) NPTORD 4500-54

Notes to Instructor
If DWAN workstations not accessible instructor to provide hard copy calendars to be filled out by hand by students. If each student has own DWAN workstation, option to have students watch the video individually and complete the forms at their own workstation.

Limitations	No face to face feedback when assessing students (done as written). Not reflective of realistic last minute ships schedule changes in training environment.
Impact of Failure	IAW NPTORD 4500-54

Item	PC 001 – PERFORM THE DUTIES OF A TRAINING PETTY OFFICER	SAT	UNSAT	Remarks
1. Plan/design Training Calendar				
All expired CRRs are scheduled		☐	☐	
Alongside CRRs scheduled during a/s period		☐	☐	
At sea CRRs scheduled during at sea time		☐	☐	
CRR opportunities are maximized		☐	☐	
Unscheduled/Scheduled CRRs are substantiated		☐	☐	
Individual training scheduled for subordinates		☐	☐	
OJPR progress review scheduled		☐	☐	
External units to book are identified		☐	☐	
Resources for personnel are identified		☐	☐	
Resources for equipment are identified		☐	☐	
Resources for expendables are identified		☐	☐	
Schedule conflict resolution is adhered to		☐	☐	
2. Correct Individual Training Deficiencies				
Individual performance deficiencies are noted		☐	☐	
Impact of performance deficiencies stated		☐	☐	
Solution to training deficiency provided		☐	☐	
Substantiation on signing off/not signing off OJPR item provided		☐	☐	

Score: / 16

Trainee must achieve 12/16 SATs overall, but MUST achieve minimum of 3/4 in section 2 to pass PC 001

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 002 – Assist in Mission/Exercise Preparations

1. **Performance**: Assist in Mission/Exercise Preparations

2. **Conditions**:

   a. Given:

      (1) Mission/Exercise;
      (2) DWAN workstation;
      (3) CSNI workstation;
      (4) References;
      (5) Oceanographic data;
      (6) Message drafting software; and
      (7) Threat matrix template.

   b. Denied: NIL

3. **Environmen**t: Any (at sea/alongside/shore facility & day/night)

4. **Standard**: The SCS shall assist in mission/exercise preparations by:

   a. Preparing UWW messages by:

      (1) Extracting mission specific information from:
         (a) OPGEN;
         (b) OPTASK ASW/OPTASK SCC;
         (c) TABORDs;
         (d) OPSTAT UNIT; and
         (e) CASEX relaxation tables (AXP 1);
         (f) COMPLANS, EMCON Plans and Frequency Plans (A0035);
      (2) Preparing UWW exercise support requests (D0045) to include:
         (a) Request For Information (RFI);
         (b) Water space management messages (UWON/RSS); and
         (c) TABORDs;

   b. Displaying Intelligence via UWW brief (B0035) by:

      (1) Disseminating Intelligence information (B0040) by:
         (a) Interpreting intelligence data;
         (b) Identifying capabilities and limitations of platforms (B0045); and
         (c) Producing a threat matrix;
      (2) Interpreting oceanographic data (F0045) by:
         (a) Determining oceanographic effect on acoustic operations

(C0150);

(b) Establishing capabilities and limitations of towed array operations in OPAREAS (C0160); and

(c) Calculating array tow speeds, tilt and/or balancing (C0120).

5. Training Limitations: NIL

EO 002.01 - Preparing UWW messages

002.01.01 – Interpreting OPGEN & OPSTAT UNIT

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
150 min	confidential	RQ-PO2 Instructor or UWWD

Learning Objectives	Method
1) General Message Format 2) Interpreting OPGEN messages 3) Interpreting OPSTAT Unit messages 4) Familiarize with OPTASK COMMS (COMPLAN, EMCON and Frequency Plan)	Walk through/talk through lecture 3 Star Student study/presentation

Activity Steps

1) Instructor to provide lesson with students following along in APP 11. (40 min) Lesson to include message hierarchy and key formatting components of messages to include:
    a. action addressees
    b. info addressees
    c. message Classification
    d. Subject line/Message ID
    e. Subject Indicator Code (SIC)
    f. references
    g. Message body
    h. Drafter/Releasing Authority completed
2) Instructor to walk through a OPTASK COMMS identifying key components related to ASW (10 min)
3) Instructor to then provide students with an OPGEN message, allowing time for them to read and interpret the general information contained in the message. (20 min)
4) Students are then asked to write down the three most important points contained in the OPGEN message. (5 min)
5) Students are to then share their responses with the class (any repeated responses can be skipped). (5 min)
6) Students are then provided 2 OPSTAT Units (1 Canadian & 1 Allied nation) and must use APP 11 to interpret the messages (25 min)
7) The class is then split in half, 1 group to prepare a presentation on content on Unit#1 and the other group to present their content on Unit#2 (using the whiteboard). They are provided time to consolidate their responses and provide the “so what” considerations to the class. (15 min).
8) Groups are then to present their unit and the application of this information to the class (10 min each/ 20 min total). Presentations to include:

	a. Units name b. Sensors c. Weapons d. Aircrafts e. Equipment limitations/deficiencies f. Caps/lims (speed, endurance, etc.) 9) Instructor to provide feedback and answer any questions on message format and extracting information from OPGENS and OPSTAT units (10 min)

Resources	References
1) CSNI computer/SECRET stand-alone (13) 2) Projector 3) OPGEN message 4) OPSTAT Unit messages (x2 per student) (CAN & Allied) 5) Whiteboard & markers	1) APP 11

Notes to Instructor

EO 002.01 - Preparing UWW messages

002.01.02 – Interpreting OPTASK ASW/SCC

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ-PO2 Instructor or UWWD

Learning Objectives	Method
1) Interpret data from OPTASK ASW(Supp) 2) Interpret data from OPTASK SCC	Talk through, walk through lecture

Activity Steps

1) Instructor to lead the class through the format for the OPTASK ASW/SCC message (using APP 11) highlighting the components of the message that would be most applicable to the SCS (building a list on the board) (30 min). List should include:
    a. EXER / OPER identifier;
    b. MSGID identifier;
    c. Period of Time
    d. SUBTHREAT
    e. CONDUCT
    f. SCREEN disposition
    g. FASWMSSN (Force ASW Shipborne Sensor and Equipment)
    h. FASWTASK
    i. TDZTDA
    j. ASTD (Anti-Ship Torpedo Defence) instructions
    k. ASW Pre-Planned Responses
    l. ASIAP (Active Sonar Interference Avoidance Plan)
    m. SIAP (Sonobuoy Interference Avoidance Plan)
    n. Acoustic Deception Plans
2) Students are then broken into pairs and provided an OPTASK ASW, OPTASK ASW Supp or OPTASK SCC. Students then asked to breakdown the message they were provided, using the list created as a class during the lesson (on the board). Students work in pairs and complete the task. (55 min)
3) Instructor to then call on students to provide their answers based on the list discussing the gaps or surplus of information found. (15 min)

Resources	References
1) CSNI computer/SECRET stand-alone (13) 2) Projector 3) OPTASK ASW message 4) OPTASK ASW Supp message	1) APP 11

5) OPTASK SCC	

Notes to Instructor

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EO 002.01 - Preparing UWW messages

002.01.03 – Interpreting TABORDs and CASEX Relaxation Tables

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
250 min	confidential	RQ-PO2 Instructor or UWWD

Learning Objectives	Method
1) Interpret TABORD messages 2) Interpret CASEX relaxation tables 3) Interpreting exercise instructions	Talk through, walk through lecture Student research/presentation

Activity Steps

1) Instructor to conduct a walk through, talk through lesson utilizing AXP 1 Table Order 3-1 and discuss the serial details covered in the message. Students are then provided an example message and instructor to lead discussion on the key information contained in the TABORD. (50 min)
2) Students are then paired up and provided a second TABORD (each group provided a different message) to break down. They are to prepare their information in order to present the class (whiteboard/flip chart). (30 min)
3) Students present the contents of their TABORD to the class. (20 min)
4) Instructor to then explain CASEX relaxation tables to the class, discussing who authorizes them and what you are able to do with each, which ones can be added without authorization (using AXP 1 Table Order 3-2). Explain the key/common relaxations that apply to CASEX. Break down the same message as used throughout step 1 (50 min)
5) Students are then asked to look back into their TABORDs and determine which relaxations are in effect/apply (using Table Order 3-2). (25 min)
6) Instructor to then explain exercise instructions to the class, which apply to subs, A/C and ships (using AXP 1 Table Order 3-3), break down the same message as used throughout step 1. (50 min)
7) Students are then asked to look back into their TABORDs and determine which exercise instructions apply (using Table Order 3-3). (25 min)

Resources	References
1) CSNI/SECRET standalone computer (13) 2) Projector 3) Example TABORDS x 8 (different messages) 4) Copy of Table Order 3-1, 3-2, & 3-3	1) AXP 1

Notes to Instructor

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EO 002.01 - Preparing UWW messages

002.01.04 – Drafting RFIs

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Utilizing the RFI website/tool 2) Determining RFI requirements 3) Drafting RFIs 4) Submitting RFIs	Guided web based lesson practical

Activity Steps	
	1) Instructor to lead students through the Trinity website in searching their website to determine if the information is already in existence. They will also discuss how to look through the RFI log to determine pending intelligence products before submitting their own. (25 min) 2) Instructor to then discuss what occasions they would need an RFI (what info they may be looking for) and how to use the RFI drafting/submission tool and what information is required to be completed. Of note: RFIs are not limited to intelligence products, but may also apply to oceanographic support products. Instructor to touch on RSS submissions (service support during a mission: ex METOC). (25 min) 3) Students are then paired up and asked to use the mission specific message traffic (OPORD) and determine their RFI requirements. Teams are to use the Trinity website to determine what products are already in existence that apply, and which information they will be required to submit and RFI for. Students are to use the soft copy of the RFI template to create their "EXERCISE" RFIs (no risk of accidental submission). (30 min) 4) Students are submit their RFI by either emailing to instructor or having instructor come to their workstations to check the accuracy. (5 min) 5) Class to then discuss what RFIs they determined were necessary to submit. (15 min)

Resources	References
1) CSNI computer (13) 2) projector 3) soft copy RFI (x12) 4) Operation specific OPORD	1) Trinity webpage 2) NAVORD 2000-2 (RFI)

Notes to Instructor
Use the Mission OPORD for the course mission.

EO 002.01 - Preparing UWW messages

002.01.05 – Drafting water space management messages

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
500 min	Up to SECRET	RQ-PO2 Instructor/UWWD

Learning Objectives	Method
1) Interpreting OPAREA Assignment message 2) Drafting RSS messages 3) Drafting UWON messages 4) Confirming receipt of acknowledgement message	Lecture practical

Activity Steps	
	1) Instructor to lead students through reading an OPAREA assignment message (indicating how to read them and what information is pertinent, reading the notes area regarding SUB ops). (15 min). 2) Instructor to identify the various water space management areas for other countries IAW AHP 6. (15 min) 3) Instructor to lead students through drafting RSS message using the CANSUBFOR lecture (found online at CANSUBFOR webpage). (50 min) 4) Students are to then use message drafter to develop a template that can be used for follow on activities. (30 min) 5) Instructor to lead students through drafting UWON messages using the CANSUBFOR lecture (online) Note that on the job SCS may be required to determine the coordinates but on course we will base scenario off UWWD to provide that information. (50 min) 6) Students are then to use message drafter to develop a template that can be used in follow on activities. (30 min) 7) Students are then provided a scenario where they would be required to draft an RSS message and a UWON message. Students may work in pairs, and instructor to provide feedback (walk around to view, print or via email). (150 min) 8) Students are then provided a secondary scenario where they are required to draft an RSS message and a UWON message. Students may work in pairs, and instructor to provide feedback (walk around to view, print or via email). (150 min)

Resources	References
1) CSNI/SECRET workstations with message drafter 2) OPAREA message	1) APP 11 2) MARPACORD 3350-1 3) MARLANTORD 3255-1

3) CANSUBFOR lectures (RSS and UWON) 4) 2 x Scenario requiring an RSS and UWON each	4) AHP 6

Notes to Instructor

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EO 002.01 - Preparing UWW messages

002.01.06 – Drafting TABORDs

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
250 min	Up to SECRET	RQ-PO2 Instructor/UWWD

Learning Objectives	Method
1) Drafting CASEX TABORDs 2) Advise on CASEX relaxations table requirements	Experiential learning (self-guided)

Activity Steps	
	1) Instructor to break class into pairs, and provide a scenario where they must draft a CASEX TABORD message. Students are to use the references and knowledge gained from previous lessons to draft 1 X CASEX TABORD message per team. (150 min). 2) Instructor to provide feedback on each teams message (hard copy or electronic based on classroom assets). 3) Students provided a new scenario and must draft a second CASEX TABORD, this time individually. Instructor to provide feedback to each student (hard copy or electronic based on classroom assets). (100 min)

Resources	References
1) CSNI/SECRET workstation with message drafter 2) Scenarios x 2 (6 copies of 1st, 12 copies of 2nd) 3) Approved Starred relaxations	1) AXP 1

Notes to Instructor

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EC 002.01 - Prepare UWW messages

Scope	Conditions	Sequence
EO 002.01	Classified classroom	O/C EO 002.01
Time	**Classification**	**Instructor**
150 total	CONFIDENTIAL/SECRET	UWWD Instructor

Scenario	Ship is preparing to conduct CASEX, RQ-PO2 student will be assisting in the exercise preparations by submitting an RSS/UWON and TABORD.
Assessment Steps	Input: Students will be provided: - Scenario - OPTASK ASW - message templates Performance: Students will have 150 min to interpret the message traffic and data provided IOT draft a TABORD and RSS/UWON Output: Each student will submit - 1x TABORD - 1x RSS/UWON Feedback: Feedback to be provided to the student using EC 002.01 checklist in written form within 48hrs.

Resources	References
1) DWAN workstation 2) CSNI workstation (or classified laptop) with Microsoft Office 3) Message drafting software 4) Scenario (including OPTASK ASW)	1) APP 11 2) MARPACORD 3350-1 3) MARLANTORD 3255-1 4) AXP 1

Notes to Instructor
RSS/UWON selection is based off the mission, not the posting of the students Instructor to determine and communicate method of submission to the students (based on classified content: CSNI email, Classified USB stick, paper copy, or instructor rotating through student workstations).

Limitations	
Impact of Failure	IAW NPTORD 4500-54

Item	EC 002.01 - PREPARE UWW MESSAGES	SAT	UNSAT	Remarks
1. Prepare water space management message (UWON/RSS)				
Required action addressees are listed		☐	☐	
Required info addressees are listed		☐	☐	
Classification identified		☐	☐	
Subject line is completed		☐	☐	
Subject Indicator Code (SIC) listed		☐	☐	
All references are included		☐	☐	
Message adheres to UWON/RSS format		☐	☐	
Message body info complies with provided exercise direction				
Exercise participants are listed		☐	☐	
OPAREA is specified		☐	☐	
Date/Time of water space requirement is specified		☐	☐	
Drafter/Releasing Authority completed		☐	☐	
2. Prepare UWW exercise message (TABORD)				
Required action addressees are listed		☐	☐	
Required info addressees are listed		☐	☐	
Classification identified		☐	☐	
Subject line is completed		☐	☐	
Subject Indicator Code (SIC) listed		☐	☐	
All references are included		☐	☐	
Message adheres to TABORD format		☐	☐	
Message body info complies with provided exercise direction				
Relaxations and special instructions are listed		☐	☐	
Safety requirements are listed**		☐	☐	
Exercise participants are listed		☐	☐	
Communications and reporting is indicated		☐	☐	
OPAREA is specified		☐	☐	
Date/Time of exercise is specified		☐	☐	
Drafter/Releasing Authority completed		☐	☐	

Score: / 25
The trainee must achieve 18/25 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 002.02 - Displaying Intelligence via UWW brief

002.02.01 – Interpreting Intelligence Data

Conditions		Sequence
TRINITY/ADAC TOUR or Classified classroom		
Time	**Classification**	**Instructor**
350 min	Up to SECRET	RQ-PO2 Instructor & Guest lecturer (if tours)

Learning Objectives	Method
1) Identify resources of intelligence data 2) Utilize intelligence resources (classified/Unclassified/open source)	Tours or Lecture T-chart/brainstorming

Activity Steps	
	1) Trinity & ADAC brief: conducted as tours of facilities or if unavailable, instructor to conduct prepared lessons. (200 min). 2) Instructor to provide a list of resources for intelligence data to the students. List to identify Trinity/classified resources, unclassified resources and open source/ trusted sources. Students are broken into 3 groups and assigned one of the sources to research. Groups are to create a T-Chart identifying the pros & cons of the resources, brainstorming what data can be found and how it can be used. (100 min) 3) Groups to present their findings to the class.(40 min) 4) Instructor to lead a wrap up identifying where they will use this information. (10 min)

Resources	References
1) DWAN workstations (6) 2) CSNI workstations (6) 3) Tour facility booking OR prepared lectures for ADAC/TRINITY 4) projector 5) list of intel resources handout 6) flipchart/whiteboard & markers	1) CFCD 106 2) Jane's Fighting Ships 3) DIAS Threat Web 4) Trinity Webpage 5) Open Source

Notes to Instructor
Students may come up with their own sources but must be critical in their reputable nature and provide bibliographical information.

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EO 002.02 - Displaying Intelligence via UWW brief

002.02.02 – Identify Capabilities/Limitations of Platforms

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
150 min	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) ID key data on friendly units 2) ID key data on threat units 3) Producing a threat matrix	Student research

Activity Steps

1) Students are provided a list of platforms (from the course mission, additional units added based on class numbers based on instructors prerogative), and must select min 1 unit to research. Students must identify the capabilities and limitations of the selected unit. (150 min) (Research must include:
    a. Name/country
    b. Emitters
    c. Weapons (torpedoes, missiles & mines)
        i. Range/speed/run time of torpedoes
        ii. Method of homing
    d. Speeds (max and best endurance)
    e. Engineering configuration
    f. Operating depth
    g. National doctrine
        i. Mission
        ii. Competency
        iii. Experience
2) Instructor to have a threat matrix template on the instructor computer. As students complete their research they are to input their data into the class threat matrix. The national doctrine component can be saved IOT be used in a future lesson (building the brief).

Resources	References
1) DWAN workstations (6) 2) CSNI workstations (6)	1) CFCD 106 2) Jane's Fighting Ships 3) DIAS Threat Web 4) Trinity Webpage 5) Open Source

Notes to Instructor

Threat Matrix will be used during PO 003
Instructor must be available to answer student questions throughout research periods

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EO 002.02 - Displaying Intelligence via UWW brief

002.02.03 – Interpret Oceanographic Data

Conditions		Sequence
classified classroom		
Time	**Classification**	**Instructor**
100 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Interpret Environmental conditions of oceanographic data 2) How environment effects ASW sensors	Guided lecture 3 square chart

Activity Steps

1) Instructor uses created "package" of oceanographic data to include:
    a. Ocean features analysis
    b. Temperature profile
    c. Salinity profile
    d. Sound velocity profile
    e. Ocean currents/fronts/eddies
    f. Bottom topography
    g. Bottom composition
    h. Sounds path/channels
        i. Cut off frequencies
        ii. Shadow zones
    i. Shipping density
    j. Biological impact
    k. Weather/Sea State
2) Students are encourage to discuss each source/slide providing the characteristics/data shown and the application/"so what" of the oceanographic data. One student is selected to create a table to capture the discussion points on behalf of the class for each slide. (100 min)

Characteristics shown	
Effect on own ships sensors	Effect on submarines operating

Resources	References
1) DWAN computer 2) Projector 3) Oceanographic "package" 4) Flipchart & markers	1) RP33 2) MILOC manual

Notes to Instructor

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EO 002.02 - Displaying Intelligence via UWW brief

002.02.04 – Determining Oceanographic Effect

Conditions		Sequence
Classified classroom		After 002.02C
Time	**Classification**	**Instructor**
250 min	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Estimate tactical oceanographic range predictions 2) Establishing caps/lims of towed array in area of operations 3) Providing sensor recommendations 4) Calculating array tow speed/tilt/balancing	Guided discussion Lecture

Activity Steps

1) Class to review oceanographic package data from previous lesson and discuss the estimated oceanographic range prediction. (100 min).
   Students are to discuss:
   a. factors influencing ranges of own ship
   b. factors influencing enemy subs
   c. own ship mission
   d. threat ship mission
   e. submarine operating depth
   f. speed of advance
   g. physical ocean limitations
   h. sound profile
2) The instructor is to then provide instruction on how to calculate tow speed/cable scope/depth and safety considerations IAW NAVORD 3137-1. (20 min)
3) The class is to then make a recommendation for how to employ ships sensors. (30 min)
4) Students are then broken into groups of 2-3, and provided a new oceanographic data package (70 min). They must then create a table based on their scenario:

Key data	
Effect on own ships sensors	
Effect on submarines operating	
Recommended sensor employment	

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	5) Students are to present their findings to the class, and instructor to provide feedback. (30 min).

Resources	References
1) DWAN/CSNI workstation 2) Oceanographic data packages (6) 3) Projector 4) Flipchart (sticky note) and markers	1) RP 33 2) MILOC manual 3) NAVORD 3137-1 4) ATP 28

Notes to Instructor
Mathematical formula for calculating tow speed: (cable scope/ships speed)/2

EO 002.02 - Displaying Intelligence via UWW brief

002.02.05 – Building Intel/Oceanographic Brief

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
300 min	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Building intelligence brief 2) Building tactical oceanographic brief	Student briefs

Activity Steps	
	1) Students are provided a template for intel/oceanographic brief, intel data (from lesson 002.02B) and oceanographic data package. They must utilize this information to individually build an intel/oceanographic brief. (100 min) 2) Students to present their briefs to the class, instructor to provide feedback on brief construction and content (not on briefing style). (16 min per student including debrief)

Resources	References
1) CSNI workstation 2) Projector 3) Intel/oceanographic brief template	1) RP 33 2) MILOC manual 3) NAVORD 3137-1 4) ATP 28 5) CFCD 106 6) Jane's Fighting Ships 7) DIAS Threat Web 8) Trinity Webpage 9) Open Source

Notes to Instructor

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EO 002.02 - Displaying Intelligence via UWW brief

EC 002.02 - Display Intelligence via UWW brief

Scope	Conditions	Sequence
EO 002.02	Classified classroom	O/C EO 002.02
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ-PO2 Instructor

Scenario	Ship is preparing to conduct CASEX, RQ-PO2 student will be assisting in the exercise preparations by submitting an intel/oceanographic brief.
Assessment Steps	Input: Students will be provided: - Scenario - RFI response - OPAREA assignment message/ UWON approval - RSS/UWON (completed for OPAREA request) - brief template - Oceanographic data (bathy, bottom composition, topography, OFA, and ranges) Performance: Students will have 100 min to interpret the message traffic and data provided IOT develop an intel/oceanographic brief. The brief must include: intelligence for platform capabilities and limitations, oceanographic data, and recommended sensor employment. Output: Each student will submit - 1x intel/oceanographic brief Feedback: Feedback to be provided to the student using EC 002.02 checklist in written form within 48hrs.

Resources	References
1) DWAN workstation 2) CSNI workstation (or classified laptop) with Microsoft Office 3) Brief template 4) Scenario (including exercise package)	1) Jane's Fighting Ships 2) Ocean Navigator 3) Trinity Web based Reference 4) Open source references

Notes to Instructor	
The briefs created will not be presented. Instructor to determine and communicate method of submission to the students (based on classified content: CSNI email, Classified USB stick, paper copy, or instructor rotating through student workstations).	
Impact of Failure	IAW NPTORD 4500-54

Item	EC 002.02 - DISPLAY INTELLIGENCE VIA UWW BRIEF	SAT	UNSAT	Remarks
1. Displaying Intelligence via UWW brief				
Platform threats are identified		☐	☐	
Speed capabilities/limitations are identified		☐	☐	
Operating depths are identified		☐	☐	
Weapons systems (caps/lims) identified - Torpedo - Missiles - mines		☐	☐	
Emitters vs own ships capabilities are identified		☐	☐	
Anticipated threat posture		☐	☐	
OPAREA identified		☐	☐	
Bottom topography identified		☐	☐	
Bottom composition indicated		☐	☐	
Sound profile indicated		☐	☐	
Sound paths indicated		☐	☐	
Cut off frequencies are identified		☐	☐	
Sonic layer depth/Sub best depth		☐	☐	
Ocean Features Analysis (OFA) is provided - fronts - currents - eddies		☐	☐	
Meteorological data provided		☐	☐	
Acoustic Ranges indicated		☐	☐	
Recommended tactical employment of sensors justified/substantiated:				
	HMS	☐	☐	
	Sonobuoy	☐	☐	
	CANTASS	☐	☐	

Score: / 19

The trainee must achieve 14/19 SATs to pass this EC.

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Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 002 - Assist in mission/exercise preparations

PC 002 - Assist in mission/exercise preparations

Scope	Conditions	Sequence
PO 002	Classified Classroom	O/C PO 002
Time	**Classification**	**Instructor**
200 min total	Up to SECRET	RQ- PO2 Instructor

Scenario	Ship is preparing to conduct CASEX, RQ-PO2 student will be assisting in the exercise preparations by submitting a TABORD and intelligence/oceanographic portion to be used in CASEX brief.
Assessment Steps	Input: Students will be provided: - Scenario - OPTASK ASW - RFI response - OPAREA assignment message/ UWON approval - RSS/UWON (completed for OPAREA request) - CASEX approved relaxations - brief template - Oceanographic data (bathy, bottom composition, topography, OFA, ranges) Performance: Students will have 200 min to interpret the message traffic and data provided IOT draft a TABORD for the CASEX and develop an intel/oceanographic brief. The brief must include: intelligence for platform capabilities and limitations, oceanographic data, and recommended sensor employment. Output: Each student will submit - 1x TABORD - 1x intel/oceanographic brief Feedback: Feedback to be provided to the student using PC 002 checklist in written form within 48hrs.

Resources	References
1) DWAN workstation 2) CSNI workstation (or classified laptop) with Microsoft Office 3) Message drafting software 4) CASEX Brief template 5) Scenario (including exercise package)	1) APP 11 2) MARLANTORD 3255-1 3) MARPACORD 3350-1 4) AXP 1 5) Jane’s Fighting Ships 6) Ocean Navigator 7) Trinity Web based Reference 8) Open source references

Notes to Instructor
The briefs created will not be presented as part of the assessment for PC 002, however will be used as an activity within PO 003. Instructor to determine and communicate method of submission to the students (based on classified content: CSNI email, Classified USB stick, paper copy, or instructor rotating through student workstations).

Limitations	Timings are not in alignment with how it would be done on ship, however this is mitigated by providing much of the information to the students and understanding on ship there attention would be split with other obligations.
Impact of Failure	IAW NPTORD 4500-54

Item	PC 002 - ASSIST IN MISSION/EXERCISE PREPARATIONS	SAT	UNSAT	Remarks
1. Prepare UWW exercise message (TABORD)				
Required action addressees are listed		☐	☐	
Required info addressees are listed		☐	☐	
Classification identified		☐	☐	
Subject line is completed		☐	☐	
Subject Indicator Code (SIC) listed		☐	☐	
All references are included		☐	☐	
Message adheres to TABORD format		☐	☐	
Message body info complies with provided exercise direction				
Relaxations and special instructions are listed		☐	☐	
Safety requirements are listed**		☐	☐	
Exercise participants are listed		☐	☐	
Communications and reporting is indicated		☐	☐	
OPAREA is specified		☐	☐	
Date/Time of exercise is specified		☐	☐	
Drafter/Releasing Authority completed		☐	☐	
2. Displaying Intelligence via UWW brief				
Platform threats are identified		☐	☐	
Speed capabilities/limitations are identified		☐	☐	
Operating depths are identified		☐	☐	
Weapons systems (caps/lims) identified - Torpedo - Missiles - mines		☐	☐	
Emitters vs own ships capabilities are identified		☐	☐	
Anticipated threat posture		☐	☐	
OPAREA identified		☐	☐	
Bottom topography identified		☐	☐	
Bottom composition indicated		☐	☐	
Sound profile indicated		☐	☐	
Sound paths indicated		☐	☐	

Cut off frequencies are identified	☐	☐	
Sonic layer depth/Sub best depth	☐	☐	
Ocean Features Analysis (OFA) is provided - fronts - currents - eddies	☐	☐	
Meteorological data provided	☐	☐	
Acoustic Ranges indicated	☐	☐	
Recommended tactical employment of sensors justified/substantiated:			
HMS	☐	☐	
Sonobuoy	☐	☐	
CANTASS	☐	☐	

Score: / 33

The trainee must achieve 25/33 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 003 – Contribute to Intelligence gathering Operations (B0030)

1. **Performance**: Contribute to Intelligence gathering Operations (B0030)

2. **Conditions:**

   a. Given:

      (1) Operations Room;
      (2) ASW Team;
      (3) References;
      (4) Command System
      (5) SHINCOM;
      (6) NIXIE; and
      (7) Mission/Exercise.

   b. Denied: NIL

   c. Environmental:

      (1) Ship (at sea/alongside & day/night); or
      (2) SEACOT/SEAWOLF trainer.

3. **Standard**: The SCS shall Contribute to Intelligence gathering Operations (B0030) by:

   a. Operating Communications Equipment (A0010) by:

      (1) Performing communications circuit functional checks (A0005);
      (2) Conducting internal/external voice reports;
      (3) Monitoring internal/external circuits;
      (4) Disseminating received communication information;
      (5) Encoding/Decoding tactical information (with authenticate communications transmissions) (A0025)
      (6) Authenticate communication transmissions (A0030); and
      (7) Performing SAR/Distress/SUBSMASH communication transmissions (A0020);

   b. Operating Command System (CS) (B0050) by:

      (1) Setting up the CS;
      (2) Maintaining Stateboards (B0015);
      (3) Upgrading track information; and
      (4) Plotting formations, screens & dispositions (B0010); and

(5) Generating tactical decision aids;

c. Coordinating the employment of UWW weapons, sensors and decoys (C0140) by:

(1) Setting up sensor equipment to include:
  (a) Towed array sonar system (C0127);
  (b) Ancillary equipment;
  (c) HMS; and
  (d) SPS;
(2) Directing ASW team with search procedures;
(3) Adhering to EMCON restrictions (B0025)
(4) Compiling ASW tactical picture (E0005) to include:
  (a) Detecting radar contacts (E0015);
  (b) Assessing UWW threat info (D0050) by:
    i. Localizing Passive Acoustic Contacts (C0085); and
    ii. Constructing contact AOP (C0100);
  (c) Conducting quality assurance of ASW picture;
  (d) Analyzing TMA data (C0095) by:
    i. Setup target motion analysis plot (C0086);
    ii. Plotting target motion analysis (TMA) data (C0090); and
    iii. Evaluating range predictions;
  (e) Evaluating correlated acoustic info (F0005);
  (f) Correlating active/passive/non acoustic info (C0105);
  (g) Identifying changes in contacts readiness tactical states (F0015); and
  (h) Amplifying track info (E0020) within;
    i. Command system; and
    ii. Internal/External voice reports
(5) Conducting Torpedo Detection and Tracking (TDT) Procedures to include:
  (a) Enforcing emission control (EMCON) restrictions (B0020);
  (b) Operating NIXIE;
  (c) Utilizing sensors for detection and tracking;
  (d) Directing sensor employment (sonobuoys);
  (e) Providing amplifying information/recommendations via voice reports; and
  (f) Providing weapon/torpedo status report.

4. Training Limitations

EO 003.01 - Operate Communications Equipment

003.01.01 – Basic Communications Theory

Conditions		Sequence
Classified Classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	Guest Lecturer MS NCI OP Instructor

Learning Objectives	Method
1) Explain basic communication frequencies ranges 2) Understand power limits and implications to operations 3) Environmental effects 4) Tactical communications	Interactive Lecture

Activity Steps	
	1) The instructor will discuss Capabilities and limitations MF/HF, UHF, VHF frequencies of example messages using visual aids. The instructor will ask questions about frequency related information throughout in order to assess student understanding of the Material (50 min). a. Frequency range; b. Power limits and implication to operations; and c. Environmental effects 2) The instructor will discuss Tactical communications and special communications: (a) Types of Nets; (b) EMCON/restrictions to communications; (c) COMSEC violations;

Resources	References
1) Example Comms state and EMCON plan 2) Blank Stateboard printouts, enough for each student 3) PowerPoint lesson	1) CFCD 106 Maritime Tactical Instructions 2) National Defence Security Orders and Directives (NDSOD) 3) NOSTP 200 - CIO Manual 4) Operator's Guide SHINCOM applicable to HALIFAX CLASSACP 125 Communications Instructions and Radio Telephone Procedures 5) ACP 176 Allied Naval and Maritime Air Communications Instructions 6) APP 1 Allied Maritime Voice Reporting 7) APP 7 NATO Maritime Code words Publication

Notes to Instructor

EO 003.01- Operate Communications Equipment

003.01.02 – Operate Internal / External SHINCOM Nets

Conditions		Sequence
SEACOT		
Time	**Classification**	**Instructor**
100 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Setup SHINCOM terminal (internal / external)	EDI
2) Demonstrate Secure / Unsecure communications	Instructor guided
3) Conduct internal / external communication checks	Practice

Activity Steps

1) The instructor will explain the required communication nets that are to be monitored and demonstrate how to setup the SHINCOM terminal for (15 min):
   a. Internal nets;
   b. External nets;
   c. P-Nets
   Students will then set up their SHINCOM terminal as per instructor's direction. Instructor will go around the class and confirm each student is correct (10 min).
2) Instructor will then explain the differences between operating on secure network vice unsecure network and demonstrate how to change between secure / unsecure nets. Students will be required to switch between secure / unsecure nets, instructor will visually confirm students completed correct task (10 min).
3) Instructor will explain and demonstrate how to conduct an internal communication check. Students will be in groups of 2-3 and conduct internal communication checks (15 min).
4) Instructor will explain and demonstrate how to conduct an external communication check. Students will be in groups of 2-3 and conduct external communication checks (25 min).
5) Student will conduct 1 x external comms check with instructor (25 min).

Resources	References
1) SHINCOM terminal	1) APP-7
2) Unit Call signs	2) ATP 1
	3) CFCD 102 TACAIDS
	4) NOSTP 200 - CIO Manual
	5) ACP 125
	6) ACP 176

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	7) APP 1 Allied Maritime Voice Reporting 8) APP 7 NATO Maritime Code words Publication 9) AMSC 608 TC

Notes to Instructor

Instructor to have students practice external comms checks in both Secure and Unsecure networks

EO 003.01 - Operate Communications Equipment

003.01.03 – Tactical Communications

Conditions		Sequence
Classified Classroom		Prior to EO 003.01B
Time	**Classification**	**Instructor**
100 min	Unclassified	MS NCI OP Guest Instructor

Learning Objectives	Method
1) Authenticate Communication Transmissions	Guided discussions

Activity Steps

1) The guest lecturer will conduct a lesson on tactical communications and net security to include (50 min):
    a. Misfit/charter (delay of changing or early changing of call signs);
    b. Spoofing;
    c. Jamming;
    d. GUERRILLA report;
    e. NUCO;
    f. Authentication;
    g. COMSEC/EEFI;
    h. Kick/Dance/Waltz; and
    i. Tactical codes/low grade codes
2) Students are then provided NUCO, challenge and reply and low grade codes publications and asked to create transmissions (one sheet for encode and another for decode) to send to a partner. Students to practice encoding and decoding as if on an unsecure net, as well as practice conducting challenge and reply, instructor to circulate and ensure everyone is capable of conducing the exercise. (25 min)
3) Instructor to then hand out an exercise sheet to each student, containing the information required to be communicated, and students are to prepare their information for transmission. Students are then each asked to conduct their transmission to the class by voice. (15 min)
4) Students are then provided a list of brevity code words to look up and memorize (these can/may be added to their TACPACs) (10 min)

Resources	References
1) Computer with projector	1) CFCD 106 Maritime Tactical Instructions
2) Communications crypto publications	2) National Defence Security Orders and Directives (NDSOD)
3) Exercise sheets	3) NOSTP 200 - CIO Manual
4) List of Code words	4) ACP 125 Communications Instructions and Radio Telephone Procedures

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	5) ACP 176 Allied Naval and Maritime Air Communications Instructions 6) APP 1 Allied Maritime Voice Reporting 7) APP 7 NATO Maritime Code words Publication

Notes to Instructor

May continue to use the code words in a team quiz format throughout the remainder of the PO to practice and get students used to using the brevity code words.

EO 003.01 - Operate Communications Equipment

003.01.04 – Voice Reports

Conditions		Sequence
SEACOT		
Time	**Classification**	**Instructor**
250 min	Up to CONFIDENTIAL	RQ-PO2 Instructor

Learning Objectives	Method
1) Utilizing call signs 2) Situation, Assessment, Recommendation (SAR) format 3) Conduct internal voice reports a. closing up b. initial c. amplifying d. lost e. equipment status report 4) Conduct external voice reports (CANTASS, sonobuoy) a. initial b. amplifying c. lost d. equipment status report 1) Monitoring nets 2) Net discipline (beadwindow) 3) QA ASW team reports (ASPO)	EDI Practical

Activity Steps	
	1) Instructor to provide a lesson explaining the learning objectives, demonstrating the reports and how to obtain the information utilized in the reports. Specific note to internal/external net policy and procedures. (100 min) 2) Instructor to provide student activity handout with INTERNAL voice report information to support practice of internal voice reports. 3) Students are then called upon by the instructor to conduct reports IAW their personal handout. (50 min) 4) Instructor to provide students activity handout with EXTERNAL voice report information and Call/Sign they are to use in various reports. Students are then called upon by the instructor to conduct reports IAW their personal handout. Students to conduct BEADWINDOW procedures when situation requires.(100 min)

Resources	References
1) Voice report handouts 2) SHINCOM 3) Activity handouts (internal and external)	1) NCPM 600 2) Operator's Guide SHINCOM applicable to HALIFAX CLASS 3) APP 7

Notes to Instructor
Students are to be paired up to respond to other students reports for internal comms. External comms activity to build in who they are speaking to (whom needs to respond)

EO 003.01 - Operate Communications Equipment

003.01.05 – SUBSAR/Distress/SUBSMASH

Conditions		Sequence
Unclassified classroom		
Time	**Classification**	**Instructor**
100 min	Unclassified	CANSUBFOR Guest lecturer or RQ-PO2 Instructor

Learning Objectives	Method
1) Communications with Subs (caps and lims) 2) Sensor employment during a SUBSAR 3) Alongside reactions as RDS 4) Red Flare Procedures	Interactive lecture Think, pair, share Flip chart review

Activity Steps	
	1) Guest lecturer (or Instructor) to provide CANSUBFOR presentation on LOs. (50 min) 2) Instructor to explain the use of AXP 1 for underwater code words, ESUS programs, etc. Students are to read Red Flare Procedures (10 min) 3) Students are then asked to pair up and make a list of the steps/procedures for Red Flare. (10 min) 4) Then class to create one list, based on the lists they created in pairs, taking note of any pieces they missed during the pairing. (5 min) 5) Student are then broken into groups of 3-4 and asked to build a list of ships responsibilities based on their readings from JTFA/P CONPLAN SUBSAR Annex 13 & 14. (15 min) 6) Groups are then to present their lists to the class, noting any differences or areas of improvement (did they miss anything). (10 min)

Resources	References
1) CANSUBFOR Website 2) DWAN workstation 3) projector 4) JTFA/P CONPLAN Annex 13 & 14 (for each group) 5) flip charts/white board & markers	1) ATP 10 Search and Rescue Procedures Manual 2) ATP/MPT 57 The Submarine Search and Rescue Manual 3) AXP 1 4) ATP 28 5) CFCD 103 6) JTFA/P CONPLAN SUBSAR

Notes to Instructor
Guest lecturer is preferable, but lesson is provided if not possible. CANSUBFOR website can be used to pull up JTFA/P CONPLAN SUBSAR

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EC 003.01 - Operate Communications Equipment

Scope	Conditions	Sequence
EO 003.01	SEACOT	O/C EO 003.01
Time	**Classification**	**Instructor**
300 min total	Up to SECRET	RQ-PO2 Instructor

Scenario	Ship is preparing to conduct CASEX, RQ-PO2 student will be setting up communications equipment, conducting internal/external voice reports, monitoring communications networks, and to be prepared for SAR/Distress/SUBSMASH procedures.
Assessment Steps	Input: Students will be provided: - Scenario to include COMPLAN - SHINCOM terminal - Voice Reports Performance: Students will have 25 mins to conduct communication system procedures, internal/external voice report and SAR/SUBSMASH procedures. Output: Each student will conduct voice procedures IAW checklist. Feedback: Feedback to be provided verbally and written to the student upon completion of the EC 003.01 checklist.

Resources	References
1) SEACOT 2) SHINCOM terminal 3) Scenario with COMPLAN 4) Voice Reports	1) APP 7 2) NCPM 600 3) Operator's Guide SHINCOM applicable to HALIFAX CLASS 4) AXP 1 5) ATP 57 6) ACP 125

Notes to Instructor
Instructor to evaluate 2 students at a time, each student would act as a unit conducting their communication reports. Students are permitted to use individually Unclassified Voice Report TACPAC's.

Limitations	The communications net is dedicated for this training and requires less net discipline.
Impact of Failure	IAW NPTORD 4500-54

Item	EC 003.01 - OPERATE COMMUNICATIONS EQUIPMENT	SAT	UNSAT	Remarks
1. Operate Communications Equipment				
Comms net set up IAW COMPLAN		☐	☐	
Comms internal checks conducted IAW ACP 125		☐	☐	
Comms external checks conducted IAW ACP 125		☐	☐	
2. Conducting internal/external voice reports				
Internal reports conducted using SAR format		☐	☐	
Conduct SERPENT Initial Contact Report		☐	☐	
Conduct SERPENT Amplifying Contact Report		☐	☐	
Conduct SERPENT Lost Contact Report		☐	☐	
Conduct POINTER Initial Contact Report		☐	☐	
Conduct POINTER Amplifying Contact Report		☐	☐	
Conduct POINTER Lost Contact Report		☐	☐	
External equipment reports conducted IAW NCPM 600		☐	☐	
Use of code words IAW APP 7		☐	☐	
Use of call signs IAW COMPLAN		☐	☐	
3. Monitoring internal/external circuits				
Quality assurance of ASPO external voice reports, direct corrections if required		☐	☐	
Monitored and acknowledged external net communications		☐	☐	
4. Performing SUBSAR/Distress/SUBSMASH communication procedures				
Direct HMS to cease or maintain transmission (include power setting) IAW AXP 1		☐	☐	
Direct UWT to be monitored		☐	☐	
Recommend recovery of towed equipment		☐	☐	
Recommend Sonobuoy deployment		☐	☐	
Identify Minimum manning requirements IAW SSO's		☐	☐	

Score: / 20

Trainee must achieve 15/20 SATs on non-critical factors to pass this EC.

Positive Feedback	Improvement Feedback

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Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 003.02 - Operate Command System

003.02.01 – EMCON Enforcement

Conditions		Sequence
Classified classroom		
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-PO2 Instructor

Learning Objectives	Method
1) Supervise Marine Mammal Mitigation Procedures 2) Enforce SIAP/ASIAP 3) Apply EMCON WRT operational ROE	Self-study 3 star Lecture collaborative problem solving

Activity Steps

1) Students are provided NAVORD 4003-6 and SEMS SOP E05 and must read the policies individually (15 min).
2) Students are then each assigned a portion of the readings to determine the three most important points covered within their readings. (5 min)
3) Students to then share their points with the class. (5 min)
4) Instructor to conduct a lesson on ROE restrictions regarding EMCON and its application to the ASW team. Lesson to carry on and discuss SIAP/ASIAP on what it is, why we employ it, demonstrate how to use it. (15 min)
5) Class then provided a scenario where they have:
    a. 3 Canadian Patrol Frigates
    b. 1 x cyclone helicopter
    c. 1x aurora fixed wing A/C
    d. Sonobuoy channels 1-99
6) Class to then develop a SIAP and ASIAP. (10 min)

Resources	References
1) DWAN computer 2) projector 3) whiteboard & markers	1) NAVORD 4003-6 2) SEMS ANNEX SOP E05A 3) ATP 28 4) CFCD 106

Notes to Instructor

Students must account for Submarine Launched One-way Tactical (SLOT) buoys when creating their allocation plan

EO 003.02 - Operate Command System

003.02.02 – Set up Command System

Conditions		Sequence
SEACOTT		
Time	**Classification**	**Instructor**
900 min	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Set up Command System to operate in SCS mode 2) Stateboard setup/maintenance 3) updating UWW Inventory in CS	EDI practical

Activity Steps	
	1) Instructor to explain, and demonstrate the process for setting up Command System, setting up stateboards and updating inventory using the checklist as guidance on the projector. (35 min) 2) Students are then asked to imitate the Command System set up. (65 min) 3) Students are then asked to set up the CS individually without prompting, using handouts as guidance with instructor providing input. (800 min)

Resources	References
1) Projector 2) Command System x 13 (SEACOT) 3) CS setup, stateboards and inventory checklists	1) CMS 330 Combat System User Manual (CSUM)

Notes to Instructor
Students are authorized to use the checklist or self-developed UNCLAS TACPACs as performance supports.

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EO 003.02 - Operate Command System

003.02.03 – Setting up Tactical Decision Aids

Conditions		Sequence
SEACOT		
Time	**Classification**	**Instructor**
3600 min	Up to SECRET	RQ- PO2 Instructor

Learning Objectives	Method
1) Setting up Tactical Decision Aids 2) Plotting formations, screens and dispositions 3) Upgrading track information	EDI practical

Activity Steps

1) Instructor to explain and demonstrate how to input and use tactical decision aids (100 min) to include:
    a. VOSIW (and temperature)
    b. Entering a Torpedo Danger Zone (TDZ)
    c. Entering limiting lines
    d. Entering Torpedo Danger Area (TDA) around subsurface track
    e. Entering expanding Furthest on Circle (FOC)
    f. Building a water space management area with compensatory allowance
    g. Inputting a screen
    h. Torpedo Run Box (TRB)
    i. Setup torpedo threat reckon (TTR)
    j. Activate TTR
    k. Setting up TMA plot
2) Students are then asked to imitate input and use of tactical decision aids. (300 min)
3) Students are then asked to input and use of tactical decision aids without prompting, using handouts as guidance with instructor providing input. (800 min)

4) Instructor to explain and demonstrate how to input formations, screens and dispositions (100 min) to include:
    a. manual subsurface track
    b. Inputting a sonobuoy
    c. Inputting line of bearing to subsurface track
    d. Building a circular Area of Probability (AOP)
    e. Building a rectangular AOP
    f. Expanding an AOP
    g. Entering a Datum
    h. Inputting reference point

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	5) Students are then asked to imitate input formations, screens and dispositions. (300 min) 6) Students are then asked to input formations, screens and dispositions without prompting, using handouts as guidance with instructor providing input. (800 min) 7) Instructor to demonstrate how to upgrade track information (100 min) to include: a. Upgrade HMS track b. Upgrade a CANTASS bearing line c. Upgrade a sonobuoy bearing line d. Enter TDA around subsurface contact e. Enter FOC & TDA around datum f. Expand FOC & TDA g. Activate water space management h. Delete contacts i. Reposition contacts 8) Students are then asked to imitate upgrading track information. (300 min) 9) Students are then asked to upgrade track information without prompting, using handouts as guidance with instructor providing input. (800 min)

Resources	References
1) SEACOT 2) projector 3) checklists	1) CMS users guide 2) ATP 28 3) CFCD 106

Notes to Instructor
Students are authorized to use the checklist or self-developed UNCLAS TACPACs as performance supports.

EO 003.02 - Operate Command System

EC 003.02 - Operate Command System

Scope	Conditions	Sequence
EO 003.02	SEACOT	O/C 003.02
Time	**Classification**	**Instructor**
150 min total	Up to SECRET	RQ-PO2 Instructor

Assessment Steps	Input: Student provided a Command System and must conduct the checklist items upon direction from instructor (checklist headings). Performance: Student to perform CS setup, formation plots and maintain/update state boards (25 min). Output: IAW EC 003.02 checklist Feedback: immediate verbal and written feedback provided IAW EC 003.02 checklist.

Resources	References
1) Command System 2) EC checklist 3) Plotting data	1) CMS 330 Combat System User Manual (CSUM)

Notes to Instructor

Students are to be assessed 2 at a time on separate consoles (25 min per student).

Instructor direction is only asking them to conduct the checklist headings (set up Command System, Stateboard setup, input plotting data, etc.)

If SEACOT unavailable, EC can be conducted in SEAWOLF however only 1 student can be assessed at one time.

Limitations	NIL
Impact of Failure	IAW NPTORD 4500-54

Item	EC 003.02 - OPERATE COMMAND SYSTEM	SAT	UNSAT	Remarks
1. Setup Command System				
Select primary role SCS		☐	☐	
Select secondary role ASPO		☐	☐	
Select Hooked Track tab		☐	☐	
Select UWW tab		☐	☐	
Enable HMS		☐	☐	
Enable CANTASS		☐	☐	
Select NIXIE tab and verify settings IAW MARTI 2120 Annex D		☐	☐	
Select Alarm type		☐	☐	
Select sensors		☐	☐	
Select Secondary Input device tab		☐	☐	
Select Sonoouy on Secondary Input		☐	☐	
Select Range Scale 16 NM		☐	☐	
2. Setup Aux Display				
Setup Global Tracks				
	1. Turn radar On	☐	☐	
	2. Select radar X band	☐	☐	
	3. Range Rings On/Off	☐	☐	
	4. History	☐	☐	
	5. Select Leaders	☐	☐	
Setup UWW tracks tab				
	1. Display	☐	☐	
	2. History	☐	☐	
	3. Select Leaders	☐	☐	
Setup EW Tracks tab				
	1. Turn on bearing tracks	☐	☐	
Setup Special Points tab				
	1. Turn on display	☐	☐	
Setup UWW Sensors				
	1. HMS settings - enable	☐	☐	
	2. CANTASS Status - enable	☐	☐	
	3. SPS Configuration – enabling interface	☐	☐	

3. Setup Data Fusion settings			
HMS, CANTASS, SPS and UWW manual	☐	☐	
4. Threat Evaluation Weapons Assignment (TEWA) Setup			
1. Select Submarine Threat of the Day	☐	☐	
2. Select Torpedo Threat of the Day	☐	☐	
3. Update UWW inventory	☐	☐	
5. Tactical Operations			
Update Magnetic Variation	☐	☐	
Setup VOSIW	☐	☐	
Setup ASW Stateboard	☐	☐	
Setup CASEX stateboard	☐	☐	
Select the Quick links for ASW, CASEX, Radio and Fleet	☐	☐	
Setup TMA Tactical Decision Aid	☐	☐	
Setup Global TMA configuration	☐	☐	
Save changes	☐	☐	
6. Input Plotting Data			
Enter manual subsurface track	☐	☐	
Input a sonobuoy	☐	☐	
Input line of bearing to subsurface track	☐	☐	
Enter a Torpedo Danger Zone (TDZ)	☐	☐	
Enter limiting lines	☐	☐	
Enter Torpedo Danger Area (TDA) around subsurface track	☐	☐	
Build a circular Area of Probability (AOP)	☐	☐	
Build a rectangular AOP	☐	☐	
Expand an AOP	☐	☐	
Enter a Datum	☐	☐	
Enter expanding Furthest on Circle (FOC)	☐	☐	
Build a water space management area with compensatory allowance	☐	☐	
Input reference point	☐	☐	

Score: / 50

The trainee must achieve 40/50 SAT's, to pass this EC.

Positive Feedback		Improvement Feedback	
Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this EC with the candidate.		**Candidate** I have been debriefed on the result of this EC.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.01 – Compile ASW Picture

Conditions		Sequence
Classified Classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
200 min	Up to SECRET	UWWD Instructor

Learning Objectives	Method
1) Submarine Roles and Missions 2) Tactical Sonar Range and Screen planning 3) ASW Picture compilation 4) generation of AOPs 5) contact correlation procedures	Student reading (instructor directed) Student delivered content Interactive lecture

Activity Steps

1) The student will be required read in entirety the following reference ATP-28 Vol I sections (100 min):
    a. Chapter One 1103 – The Opponent
    b. Chapter 3 0318 – SONAR Detection Ranges
    c. Chapter 8 – ASWC Specified factors in ASW Defence of a Main Body
        i. 0801 -
        ii. 0802 – Theoretical Factors in ASW defence
        iii. 0803 – Limiting Lines of Approach
        iv. 0804 – Look Zone
        v. 0805 – Shadowing Zone
        vi. 0806 – Application of Theoretical Principles
    d. Annex 041 – Effects on the Environment
2) Students are then assigned topics (from the list below) to become more familiar on and present it to the class. (25 min)
3) Students are then to present their topic to the class, with instructor probing of questions regarding their topic. (40 min)
4) Instructor to lead an interactive lecture/table top of the following content: to ensure all pertinent points are covered (35 min)
    a. Submarine Roles and types of mission employment;
    b. Predicted and Tactical Sonar Ranges;
    c. Screen development (ship & helo sectors);
    d. Plot formations screens and dispositions;
    e. Anti-submarine warfare screening techniques;
    f. Torpedo Danger Zone (TDZ);
    g. Limiting lines of submerged approach;
    h. Limiting lines of quiet approach;
    i. Point Contacts vs. Line of Bearing Contacts;

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	j. Torpedo Danger Area (TDA); k. Datum l. Furthest On Circle and TDA (expansion); m. Area of Probability (Circular and Rectangular); n. AOP expansion rate; o. POD (size and expansion rate) p. Contact correlation procedures q. Reference Points

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard & markers 4) PowerPoint Lecture	1) ATP-1 2) ATP-28 3) CFCD 106

Notes to Instructor

EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.02 – Tactical Employment of ASW sensors

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
150 min	Up to SECRET	UWWD Instructor

Learning Objectives	Method
1) Tactical employment of towed arrays 2) Single ship vs multi ship procedures 3) Sonobuoy employment 4) Active sensors	Student reading Interactive lecture Scenario based collaborative problem solving

Activity Steps	
	1) The student will be required read the following references and sections, then discuss the topics point by point (100 min): a. ATP-28 0321 - Factors affecting the Tactical use of Active Sonar; b. ATP-28 0322 - Factors affecting the Tactical use of Passive Sonar; c. ATP-28 Annex A-53 – Passive Towed Array; d. ATP-28 0308 – TAS Planning Factors; e. ATP-01 12207 – Stationing of Towed Array units; f. ATP-01 12208 Towed Array Screen Station options; g. ATP-01 12209 – Towed Array Zero PIM Operations; h. ATP-28 0410 Sonobuoys; i. ATP-28 4A01 – Tactical Employment of Passive Sonobuoys; j. ATP-28 0306 – Active Sonar; and k. CFCD 106 MARTI 2160 – Littoral ASW 2) Instructor to lead an interactive lecture/table top of the following content providing scenarios to the students where they would determine how to place sensors for optimal detection: a. Tactical employment of Towed array; b. Procedures for operations with single ship towed arrays; c. Procedures for operations with multiple ship towed arrays; d. Sonobuoy employment; e. Active sensors; f. Tactical implications of littoral UWW on weapons and sensors. 3) Guest lecture opportunity to discuss LFA (50 min)

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard & markers 4) LFA lecture	1) ATP-1 2) ATP-28 3) CFCD 106

Notes to Instructor

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.03 – Command Classification

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
100 min	Up to SECRET	UWWD Instructor

Learning Objectives	Method
1) Contact classification 2) Command classification 3) DAMASS 4) Threat levels	Student reading 4 square (T-chart) Interactive lecture Case study

Activity Steps

1) The student will conduct readings to include: (30 min)
   a. ATP-01 7130 – Elements of Picture Compilation Plan;
   b. ATP-01 3770- Threat Warnings;
   c. ATP-01 12302-12303 and Figure 12-2 – Contact Classification;
   d. CFCD 120 Pg. 6-12 – Submarine Damage Assessment

2) Students are broken into two groups. Students are then to complete a 4 square chart explaining the criteria for each of the following (10 min) and each group to present their findings (5 min) with examples.

Friendly	Neutral
• Known/Allied • Ex. HMCS CALGARY	• merchant • non participatory in operations • Ex. Fishing Vessel
Suspect	**Hostile**
• unknown • potential hostile • Ex. VOI	• Conducted an attack/shows intent to attack • Combatant Enemy Ship

3) Instructor to then lead an interactive lecture on ship/force threat levels (white, yellow & red). (5 min)
4) Students are broken into groups of 2-3 and assigned a case study. Students must read the case study and brief the class on “WHAT IS THE COMMAND CLASSIFICATION, and why” (25 min)

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	5) Groups are then provide another case study to read and determine the DAMASS, to present to the class. (25 min)

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Flip charts & markers 4) Command classification Case study (5 topics, 1 per group) 5) DAMASS case study (7 topics, 1 per group)	1) ATP-1 2) ATP-28 3) CFCD 106

Notes to Instructor

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.04 – Water space management

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
50 min	Up to SECRET	UWWD

Learning Objectives	Method
1) Submarine Support Roles 2) Submarine Water space management areas 3) PMI principles and regulations 4) Reporting/communications procedures for WSM	Student reading Interactive lecture

Activity Steps

1) The student will be required read the following references and sections, then discuss the topics point by point (50 min):
    a. ATP-01 2190 – Submarine Support Operations;
    b. ATP-01 6401, 6402, 6434,6461 – 6164 – Water Space Management;
    c. ATP-01 4170 – 4173 – Submarine Communications; and
    d. AHP-6 Chapter 2 Table 4-1 & 4-2 – Permanent Submarine Exercise Areas.

2) Instructor to lead an interactive lecture/table top of the following content:
    a. Friendly Submarine Role and employment;
    b. Water space management areas;
    c. PMI principles and regulations;
    d. WSM reporting/communication procedures.

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard & markers 4) CANSUBFOR brief	1) AHP-6 2) MARLANTORD 3255-1 3) MARPAC ORD 3350-1 4) ATP-01

Notes to Instructor

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.05 – Air Operations

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
100 min	Up to SECRET	Guest Instructor: SAC or UWWD

Learning Objectives	Method
1) Rotary Wing operations 2) Fixed Wing operations	Interactive lecture

Activity Steps	
	1) The instructor to present content on Air Operations and support as applies to: a. Rotary Wing operations (caps and lims): (100 min) i. Airplans; ii. Active Dipping Sonar; iii. Sonobuoy processing capability; iv. Sonobuoy deployment; v. visual recognition (IR); vi. submarine communication/warning; vii. link capability b. Fixed Wing Operations (caps and lims); i. Airplans; ii. Sonobuoy processing capability; iii. Sonobuoy deployment; iv. visual recognition (IR); v. submarine communication/warning; vi. Magnetic Anomaly Detection (MAD); vii. link capability

Resources	References
1) PowerPoint lecture 2) CSNI/SECRET workstation 3) Projector 4) Whiteboard with markers	1) CFCD 106 MARTI's 1000 Annex C & D 2) MARTI/TACNOTE 11100, 11200, 11210, 11220, 11225, 11240,11245 3) MARTI/TACNOTE 12100, 12210, 12214, 12220, 12230, 12270

Notes to Instructor
Ensure students are aware of the new CYCLONE doctrine in Chapter 11

EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.06 – ASW Search and Attack Plans

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
100 min	Up to SECRET	UWWD

Learning Objectives	Method
1) SAU procedures 2) Search and Attack plans 3) Search techniques	Student readings

Activity Steps	
	1) Students will conduct individual readings to include (30 min): a. ATP-28 0930 – Detaching ASW Search and Attack Unit(s); b. ATP-01 6500-6502 & Table 6-15 - 6-19 – ASW Airplans, Attack, Support Plans 2) Instructor to explain Basic Search and Attack unit procedures. (20 min) 3) Students are then provided ASW search and/or attack plans and must present to the class (20 min) 4) Students are then put into groups of 2-3 and provided an area to conduct a search where they must present their search plan to the class. (30 min)

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard with markers 4) Search area activity	1) ATP-28 2) ATP-01

Notes to Instructor

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.07 – ASW Pre-Plans

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
50 min	Up to SECRET	UWWD

Learning Objectives	Method
1) ASW Pre-Plans considerations 2) Development of ASW Pre-Plan	Student readings Table Top

Activity Steps	
	1) The student will conduct individual readings on reference CFCD 106 MARTI 2550 ASW Pre-Plans (15 min). 2) Instruct and students will conduct a Table Top discussion on ASW Pre-Plan reactions to include: (35 min) a. Sonar Contact inside TDZ; b. Single ESM intercept (Surface Ship); c. Single ESM intercept (Air Asset); d. ESM Crossfix; e. TAS Bearing (single and multiple); f. Observed Sub-surface missile launch; g. Torpedo Detection.

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard with markers	1) CFCD 106 MARTI 2550

Notes to Instructor
Instructor to use references to do a talk through walk through of information necessary

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.08 – Surface and A/C attacks

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
50 min	Up to SECRET	UWWD

Learning Objectives	Method
1) Surface ship attacks 2) Aircraft attacks 3) Weapon firing picture compilation	Interactive lecture

Activity Steps	
	1) The instructor will conduct an interactive lecture (50 min) on Surface Ship & Aircraft Weapon firing criteria and procedures to include: a. Gyro Angle Snake Search (COBRA) attack b. Aircraft weapons drops (BLOODHOUND); c. Torpedo Run Box generation (DOGBOX);

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard with markers 4) PowerPoint lecture	1) ATP-28 article 0324 2) CFCD 106 MARTI 2600 3) CFCD 106 TACNOTE 2650

Notes to Instructor

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.09 – Anti-Ship Torpedo Defence

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
50 min	Up to SECRET	UWWD Instructor

Learning Objectives	Method
1) Anti-ship torpedo defence procedures 2) Evasive steering 3) TCMs	Interactive lecture

Activity Steps	1) Instructor to conduct interactive lecture on Anti-Ship Torpedo Defence procedures and requirements (50 min) to include: a. TADPOLE / Anti-Ship Torpedo Defence postures; b. Evasive steering; c. Tactical Torpedo Counter Measures (TCMs);

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard with markers 4) PowerPoint lecture	1) ATP-28 Article 1009 – 1011, Tables 10-3 to 10-7 2) CFCD 106 MARTI 2120 3) ATP 3 Chapter 1 101,103,104 & Chapter 3

Notes to Instructor
Instructor to quickly demonstrate the Screen Plans in ATP 3 Chapter 3

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.10 – IUSS & Theatre ASW

Conditions		Sequence
Classified classroom		Prior to EO 003.02
Time	**Classification**	**Instructor**
100 min	Up to SECRET	Guest Lecture: IUSS member or UWWD & Post ORO Qual NWO

Learning Objectives	Method
1) IUSS capabilities & scope of operations 2) Theatre ASW roles and responsibilities	Guest lectures

Activity Steps	
	1) IUSS guest lecturer to present on IUSS capabilities and scope of operations (to include fixed sensors and SURTASS units). Students provided an opportunity to ask questions IOT understand IUSS role. (50 min) 2) Post ORO NWO to present on theatre ASW roles and responsibilities, students provided opportunity to ask questions. (50 min)

Resources	References
1) CSNI/SECRET workstation 2) Projector 3) Whiteboard with markers 4) PowerPoint lecture IUSS 5) PowerPoint lecture Theatre ASW	1) NOPFWI IUSS Brief

Notes to Instructor

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.11 – Perform duties of SCS during CASEX

Conditions		Sequence
SEACOT preferred SEAWOLF		
Time	**Classification**	**Instructor**
350 min per student 4200 min total	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Perform the duties of SCS during CASEX 2) Ensuring towed array sonar setup 3) Directing ASW team search procedures 4) Compiling ASW tactical picture 5) Detecting radar contacts 6) Assessing UWW threat 7) Localizing passive acoustic contacts 8) Constructing AOP 9) Plotting and Analyzing TMA data 10) Correlating active/passive non acoustic info 11) Identifying changes in contacts readiness and tactical states 12) Amplifying track information	Practical Group Learning Tracking Board

Scenario	Students are to prepare CS and Coordinate the employment of UWW weapons, sensors and decoys IAW with prescribed CASEX scenarios
Activity Steps	1) Instructor to lead students through various CASEX scenarios, each student will have :CASEX A3: 2 x 50 min runs per student C2: 2 x 50 min runs per student C4: 2 x 50 min runs per student 2) Instructor to provide copy of EC 003.03A checklist for students to know the expectations while conducting their practice. 3) Students to add their “lessons learned” from their run to the class learning tracking board.

Resources	References
1) SEACOT 2) ASW team 3) student created TACPAC 4) Scenario (OPTASK ASW, TABORD)	1) AXP 1 2) ATP 28 3) CFCD 106 4) ATP 1 5) APP 7

	6) MILOC manual 7) RP 33 8) NCPM 600

Notes to Instructor

Each run is 50 min plus 10 min debrief. Students are to build a board of “lessons learned” in the trainer so each student can learn from each run even when not participating.

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EC 003.03.01 - Coordinate the employment of UWW weapons, sensors and decoys

Scope	Conditions	Sequence
EO 003.03 CASEX	SEACOT preferred SEAWOLF	O/C EO 003.03
Time	**Classification**	**Instructor**
100 min per student 1200 total	Up to SECRET	RQ-PO2 Instructor

Scenario	Students are to prepare CS and Coordinate the employment of UWW weapons, sensors and decoys IAW with prescribed CASEX scenario
Assessment Steps	Input: Students are provided a CASEX C2 scenario and support manning ASW team (IAW NWTP Vol 2) Performance: Students to prepare CS and coordinate employment of UWW weapons, sensors and decoys IAW EC 003.03A checklist. Output: IAW EC 003.03A checklist Feedback: immediate verbal and written feedback provided IAW EC 003.03 checklist.

Resources	References
1) SEACOT 2) ASW team 3) student created TACPAC 4) Scenario (OPTASK ASW, TABORD)	1) AXP 1 2) ATP 28 3) CFCD 106 4) ATP 1 5) APP 7 6) MILOC manual 7) RP 33 8) NCPM 600

Notes to Instructor
Students are authorized to use self-prepared UNCLASS TACPAC Assessment of students is 1 at a time, remainder of students may be used to man secondary roles as required (not to impact student assessment)

Limitations	Compressed timings
Impact of Failure	IAW NPTORD 4500-54

Item	EC 003.03 A - COORDINATE THE EMPLOYMENT OF UWW WEAPONS, SENSORS AND DECOYS	SAT	UNSAT	Remarks
1. Conduct Watch Preparations to include:				
Configure multi-function workstation for SCS operations - Radar - TACPLOT - Threat - Stateboards selected - CASEX stateboard annotated		☐	☐	
Setup SHINCOM IAW COMPLAN		☐	☐	
Conduct Closing up report Direct Sensor Operators team		☐	☐	
2. Supervise Sensors/Decoy employment:				
Control sensors/decoys IAW EMCON plan		☐	☐	
Adhere to ASIAP		☐	☐	
Adhere to SIAP		☐	☐	
Direct sensor settings to optimize for detect, track and localize		☐	☐	
3. Coordinate Internal Actions:				
Communicate effectively with Team (ASWC, ASPO, SAC, OOW, ORS, and ASW sensor operators) throughout		☐	☐	
Quality control the ASW related data entered into CS to include POD, AOP, DATUM, FOC, TDA		☐	☐	
Provide SITREPs to ASW sensor team		☐	☐	
Disseminate equipment casualties/ degradations to include: - Equipment - Impact - Estimated time of repair		☐	☐	
Upgrade/update contacts and system tracks (upgrade brg, track, local system track, link, and mine symbology)		☐	☐	
Localize passive acoustic contacts		☐	☐	
Construct Area of Probability		☐	☐	
Correlate contacts to include: Datum, active, passive, non-acoustic information		☐	☐	
conduct internal contact reports (Situation/Assessment/Recommendation) (TRK#, command classification, CMS picture compilation, Tactical input)		☐	☐	
4. Conduct Target Motion Analysis				
Set up TMA based on CANTASS detection		☐	☐	

Review TMA Plot	☐	☐	
Provide Assessment of TMA accuracy to ASWC	☐	☐	
Conduct TMA amplifying reports	☐	☐	
5. Coordinate External Communications:			
Conduct Initial contact reports (HMS, SPS, CATAS)	☐	☐	
Conduct amplifying reports (Dynamic events and all tactical info)	☐	☐	
Conduct Lost contact reports (HMS, SPS, CATAS with tactical consideration)	☐	☐	
Conduct equipment fault report	☐	☐	
6. Actions in Own Ship/Force Weapon Firing:			
Inform sensors team of weapon firing	☐	☐	
Enter TRB (as applicable)	☐	☐	
Direct sensor operators to conduct weapon firing procedures IAW MARTI 2122	☐	☐	
7. Green Grenade Procedures :			
Raise the alarm	☐	☐	
8. Conduct Red Flare Procedures:			
Direct HMS to cease or maintain transmission (include power setting) IAW AXP 1	☐	☐	
Direct UWT to be monitored	☐	☐	
Confirm position is marked in CMS	☐	☐	
Equipment consideration to include, recommendation of retrieval/safe distance.	☐	☐	
Recommend Sonobuoy deployment	☐	☐	
Identify Minimum manning requirements IAW SSO's	☐	☐	

Score: / 34

The trainee must achieve 26/34 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.12 - NIXIE

Conditions		Sequence
Classified classroom 003.03L (a) NIXIE laptop/trainer 003.03L(b)		
Time	**Classification**	**Instructor**
100 min lesson (003.03.12.1) 300 min (including travel time) at trainer (003.03.12.2) 400 min total	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Setup NIXIE 2) Operate NIXIE 3) NIXIE settings 4) NIXIE caps and lims	Student self-reading video Team Quiz EDI

Activity Steps	
	1) Students are required to read CFCD 106 MARTI 2120 Annex D, creating 3x questions per person to use in a team quiz activity and submit to the instructor (with answers). (50 min). 2) Students to watch a video on operating NIXIE while instructor sorts questions (15 min) 3) Instructor to divide the class into two teams, and ask the questions that were created by the students for team quiz. (35 min) 4) Travel to NIXIE trainer (if require) (15 min) 5) Instructor to conduct and EDI lesson on setting up and operating NIXIE using EC 003.03B checklist. (30 min) 6) Students provided opportunity to practice setting up and operating NIXIE IAW EC 003.03B checklist. (140 min) 7) Travel back to base (15 min)

Resources	References
1) NIXIE trainer with laptops 2) Scenario with NIXIE settings 3) CSNI/SECRET workstation 4) NIXIE video 5) projector	1) NIXIE operator manual 2) NAVORD 3137-1 3) CFCD 106 MARTI 2120 Annex D

Notes to Instructor
Instructor to make arrangements for necessary booking of trainer AND transport.

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EC 003.03.02 - Coordinate the employment of UWW weapons, sensors and decoys (NIXIE)

Scope	Conditions	Sequence
003.03 NIXIE	NIXIE laptop/trainer	O/C 003.03L NIXIE
Time	**Classification**	**Instructor**
10 min per student (including debrief) plus travel time 200 min total (including travel time)	Up to SECRET	RQ-PO2 Instructor

Assessment Steps	Input: Students asked to setup and operate NIXIE, based on provided scenario Performance: IAW EC 003.03B checklist Output: IAW EC 003.03B checklist, based on provided scenario Feedback: immediate verbal and written feedback to be provided IAW EC 003.03B checklist.

Resources	References
1) NIXIE trainer with laptops 2) Scenario with NIXIE settings	1) NIXIE operator manual 2) NAVORD 3137-1 3) CFCD 106 MARTI 2120 Annex D

Notes to Instructor
Review CFCD 106 MARTI 2120 Annex D for NIXIE settings

Limitations	Simulated equipment using laptops No external sensor to confirm NIXIE is operating
Impact of Failure	IAW NPTORD 4500-54

Item	EC 003.03B - COORDINATE THE EMPLOYMENT OF UWW WEAPONS, SENSORS AND DECOYS (NIXIE)	SAT	UNSAT	Remarks
1. Setup/Operate NIXIE				
Confirm Flash up main power unit (WENG)		☐	☐	
Flash up remote unit		☐	☐	
Choose mode of operation		☐	☐	
Select time for cycle timers		☐	☐	
Adjust time for cycle timers		☐	☐	
Turn on cycle timers		☐	☐	
Select notch filter		☐	☐	
Change notch filter		☐	☐	
Turn on notch filter		☐	☐	
Turn on Magnetic Countermeasure power		☐	☐	
*** Transmit on NIXIE**		☐	☐	
Cease transmitting on NIXIE		☐	☐	
Shutdown remote unit		☐	☐	
Shutdown Main unit in proper sequence		☐	☐	

Score: / 13 CRITIAL /1

Trainee must achieve 10/13 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.13 – Torpedo Detect and Track

Conditions		Sequence
classified classroom		
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Intro to Torpedo Detect and Track procedures/doctrine 2) TDT Picture compilation 3) Ops Team Responsibilities during TDT 4) sensor placement 5) Command guidance and reporting of TDT info 6) initial response & voice procedures 7) tracking techniques	Student self-reading guided lecture table top

Activity Steps

1) Students are required to read MARTI 2122 individually (50 min).
2) Instructor to then lead an interactive lecture, walking through the MARTI, dissecting it step by step as a group to ensure each piece it touched on. Then discussing voice procedures, initial response requirements and post attack criteria. (50 min)
3) Instructor to lead table top activity on the whiteboard, based on TDT procedures to include: (200 min)
    a. Contact position known
    b. Contact position unknown (un-alerted)
    c. Single line of bearing
    d. Bearing crossfix
    e. Datum
    f. AOP
    g. Reactions outside of TDA
    h. Reaction to consort/other units call of Torpedo

Resources	References
1) CSNI/SECRET workstation 2) projector 3) Power point lesson 4) whiteboard & markers	1) ATP 28 2) APP 7 3) CFCD 106 MARTI 2122 4) NCPM 600

Notes to Instructor

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EO 003.03 - Coordinate the employment of UWW weapons, sensors and decoys

003.03.14– Torpedo Detect and Track

Conditions		Sequence
SEACOT		
Time	**Classification**	**Instructor**
250 min per student 3000 min total	SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Torpedo Detect and Track procedures	Practical Group Learning Tracking Board

Scenario	Students will be conducting the duties of SCS within operational scenario where the ship must react to torpedo threats.
Activity Steps	1) Student provided scenario where they are conducting the duties of SCS within an ASW threat environment IAW EC 003.03C checklist. 2) Each student to be provided 5x 50 min runs (including debrief time). Students to add to the group learning board after each run to share their lessons learned. Each run will include a minimum of: two SALVOs containing two torpedoes each to track

Resources	References
1) SEACOT trainer 2) ASW team IAW NWTP Vol 2	1) ATP 28 2) APP 7 3) CFCD 106 4) NCPM 600

Notes to Instructor
Each run will include a minimum of: two SALVOs containing two torpedoes each to track (Students have option to create and use own unclassified TACPAC).

EC 003.03.03 - Coordinate the employment of UWW weapons, sensors and decoys (TDT)

Scope	Conditions	Sequence
EO 003.03	SEACOT	O/C EO 003.03
Time	**Classification**	**Instructor**
50 min per student including debrief 650 min total	SECRET	RQ-PO2 Instructor

Scenario	Students will be conducting the duties of SCS within operational scenario where the ship must react to torpedo threats.
Assessment Steps	Input: Student provided scenario where they are conducting the duties of SCS within an ASW threat environment. Performance: student to conduct TDT procedures IAW EC 003.03C checklist **Each run will include a minimum of: two SALVOs containing two torpedoes each to track (Students have option to create and use own unclassified TACPAC).** Output: IAW EC 003.03C checklist Feedback: immediate verbal and written feedback provided using EC 003.03C checklist.

Resources	References
1) SEACOT trainer 2) ASW team IAW NWTP Vol 2	1) ATP 28 2) APP 7 3) CFCD 106 4) NCPM 600

Notes to Instructor
One student to be assessed at a time. Each run will include a minimum of: two SALVOs containing two torpedoes each to track (Students have option to create and use own unclassified TACPAC).

Limitations	Stress of real torpedo attack is not felt during trainer simulation
Impact of Failure	IAW NPTORD 4500-54

Item	EC 003.03C - COORDINATE THE EMPLOYMENT OF UWW WEAPONS, SENSORS AND DECOYS (TDT)	SAT	UNSAT	Remarks
1. Conduct Torpedo Detection and Tracking				
Determine torpedo detection and confirm bearing (sensor or unit)		☐	☐	
*** * Raise the torpedo alarm (bearing from ownship)**		☐	☐	
Activate NIXIE IAW CFCD 106		☐	☐	
Adhere to ACCON policy		☐	☐	
Direct deployment of TCM sonobuoys		☐	☐	
Enter TFP for undetected contacts IAW MARTI 2122		☐	☐	
Enter TFP for Detected contacts IAW MARTI 2122		☐	☐	
Direct ASPO to pass TFP report externally		☐	☐	
Confirm input of Torpedo Threat Reckon		☐	☐	
Build TDT tactical picture IAW MARTI 2122		☐	☐	
Track torpedo(es) by generating reference point based on crossfix(es)/single line bearing		☐	☐	
Establish CMS Torpedo Tracks on Crossfix(es), direct ASPO to update and link out Torpedo Tracks when Torpedo(es) direction is determined		☐	☐	
Give SAR reports on Torpedo(es) status to command		☐	☐	
*** * Provide recommendation upon change in Torpedo status**		☐	☐	
Clean up TDT picture and refocus the ASW sensor team upon direction from command		☐	☐	

Score: / 13 CRITICALS /2

Trainee must achieve 10/13 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this EC with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 003 Contribute to Intelligence gathering Operations

003 – Consolidation: Perform the duties of SCS during Operations

Conditions		Sequence
SEACOT (preferred) SEAWOLF (secondary)		Before PC 003
Time	**Classification**	**Instructor**
6000 min total	Up to SECRET	RQ-PO2 Instructor

Learning Objectives	Method
1) Perform duties of SCS during Operations	Practical Group Learning Tracking Board

Scenario	Prepared operational mission
Activity Steps	1) Students will conduct 5 x 90 min scenario runs (plus 10 min debrief) where they will be required to set up the Command System, operate comms equipment, and coordinate the employment of the weapons, sensors and decoys in order to contribute to the intelligence gathering operations. 2) Runs to include: a) Contribute to MIO and RMP (COI and VOI); b) Covert Passive Operations (Towed Array & Sonobuoys); c) Overt Operations, Warning Process & ROE (may include TDT); d) Overt Operations Close in ASW (active sensors) and protection of a Screened unit (may include TDT); and e) SAU procedures, Detect, Track, Identify and Attack (Torpedo Detect and Tracking).

Resources	References
1) SEACOT/SEAWOLF Trainer 2) Support Manning IAW NWTP Vol 2 3) Operational mission scenario 4) NIXIE (or simulation) 5) OPTASK ASW	1) APP 11 2) ATP 1 3) ATP 28 4) APP 7 5) CFCD 106 6) MILOC manual 7) RP 33 8) NCPM 600

Notes to Instructor
Students may use self-prepared Unclassified TACPACS

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PC 003 - Contribute to Intelligence gathering Operations

Scope	Conditions	Sequence
PO 003	SEACOT (preferred) SEAWOLF (secondary)	O/C PO 003
Time	**Classification**	**Instructor**
3x 100 min per student 3600 min total	Up to SECRET	RQ-PO2 Instructor

Scenario	Prepared operational mission
Assessment Steps	Input: Students will be provided the scenario (week 1 of course) to be familiar with. Performance: Students will conduct 3 x 90 min scenario runs where they will be required to set up the Command System, operate comms equipment, and coordinate the employment of the weapons, sensors and decoys in order to contribute to the intelligence gathering operations. Students must receive a SAT on 2/3 runs. **Each run will include a minimum of: one submarine, two torpedoes to track, one contact where they must conduct TMA and at least 2 opportunity to conduct an external voice report. (Students have option to create and use own unclassified TACPAC).** Output: Students will conduct a watch IAW PC 003 checklist for assessment. Feedback: Students will be provided immediate verbal and written feedback using the PC 003 checklist.

Resources	References
1) SEACOT/SEAWOLF Trainer 2) Support Manning IAW NWTP Vol 2 3) Operational mission scenario 4) NIXIE (or simulation) 5) OPTASK ASW	1) APP 11 2) ATP 1 3) ATP 28 4) APP 7 5) CFCD 106 6) MILOC manual 7) RP 33 8) NCPM 600 9) ACP 125

Notes to Instructor
NIXIE may need to be simulated due to equipment limitations. Each student will conduct 3x 90min runs and must pass a minimum of 2/3 runs. Students will not be used in support manning roles for assessment, IAW NWTP Vol 2 Each run will include a minimum of: - one submarine - two torpedoes to track - one contact where they must conduct TMA on CANTASS - 2 opportunity to conduct an external voice report. Students have option to create and use own unclassified TACPAC

Limitations	NIXIE is not integrated and must be simulated. Inability to simulate competing for UWWD attention (prioritization of time) Net not realistic to ship/ops environment. Timings are compressed IOT assess all tasks within 90 min run.
Impact of Failure	IAW NPTORD 4500-54

Item	PC 003 - CONTRIBUTE TO INTELLIGENCE GATHERING OPERATIONS	SAT	UNSAT	Remarks
1. Operate Communications Equipment				
Comms net set up IAW COMPLAN		☐	☐	
Comms checks conducted IAW ACP-125		☐	☐	
Equipment check conducted with ASW team		☐	☐	
Voice procedures conducted IAW NCPM 600, COMPLAN & APP 7 to include:				
	Closing up report	☐	☐	
	Use of code words	☐	☐	
	Use of call signs	☐	☐	
Net discipline maintained		☐	☐	
Internal reports conducted using SAR format		☐	☐	
Sitreps provided to ASW sensor operators		☐	☐	
External reports conducted IAW NCPM 600 to include: - Initial - Amplifying (if applicable) - Lost contact reports		☐	☐	
External equipment reports conducted IAW NCPM 600		☐	☐	
Quality assurance of ASW data and reports completed IAW Preplans (OPTASK)		☐	☐	
Monitored and acknowledged external net communications		☐	☐	
Disseminated information to include: weapons firing, equipment casualties, gain/loss/change in contacts, and intelligence		☐	☐	
Communicate effectively with Team (ASWC, ASPO, SAC, OOW, ORS, and ASW sensor operators) throughout		☐	☐	
2. Operate Command System				
Setup CS with SCS primary, ASPO secondary		☐	☐	
Setup stateboards and inventory list IAW preplans and ASW briefs		☐	☐	

Setup Tactical Decision Aids IAW OPTASK ASW	☐	☐	
Confirm sensors are enabled to include:			
HMS	☐	☐	
SPS	☐	☐	
CANTASS	☐	☐	
NIXIE settings IAW preplans	☐	☐	
Confirm screens, WSM, Tactical Decision Aids and Special Points input is IAW mission/exercise	☐	☐	
3. Coordinate the employment of UWW Weapons and Sensors			
Direct sensors and decoy employment to include: setup and search orders	☐	☐	
Direct sensor settings to optimize for detection	☐	☐	
Adhere EMCON policy	☐	☐	
Adhere to SIAP / ASIAP	☐	☐	
Detect, Track and Localize passive acoustic contact	☐	☐	
Upgrade/update contacts and system tracks (upgrade brg, track, local system track, link, and mine symbology)	☐	☐	
Correlate contacts to include: Datum, active, passive, non-acoustic information	☐	☐	
Quality control the ASW related data entered into CS to include POD, AOP, DATUM, FOC, TDA	☐	☐	
Recommend Command Classification IAW ATP-01	☐	☐	
Set up TMA based on CANTASS detection	☐	☐	
Review TMA Plot	☐	☐	
Provide Assessment of TMA accuracy to ASWC	☐	☐	
Conduct amplifying reports (dynamic events)	☐	☐	
4. Conduct Torpedo Detection and Tracking			
Determine torpedo detection and confirm bearing (sensor or unit)	☐	☐	
*** * Raise the torpedo alarm (bearing from own ship)**	☐	☐	

Activate NIXIE IAW CFCD 106	☐	☐	
Adhere to ACCON policy	☐	☐	
Direct deployment of TCM sonobuoys	☐	☐	
Confirm input of Torpedo Firing Point	☐	☐	
Confirm input of Torpedo Threat Reckon	☐	☐	
Build TDT tactical picture IAW MARTI 2122	☐	☐	
Track torpedo(es) by generating reference point based on crossfix(es)/single line bearing	☐	☐	
Confirm Torpedo Tracks positions are updated linked out	☐	☐	
Give SAR reports on Torpedo(es) status to command	☐	☐	
*** * Provide recommendation upon change in Torpedo status**	☐	☐	
Clean up TDT picture and refocus the ASW sensor team upon direction from command	☐	☐	

Score: / 36 (section 1-3) /11 (section 4) CRITICALS /2 (section 4)

Trainee must achieve 27/36 SATS for sections 1-3 and 9/11 SATS for non-critical factors and ALL critical factors for section 4 to achieve PO 003

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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CHAPTER 3: ASSESSMENT OF TRAINEE

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GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of PRACTICAL or WRITTEN assessment and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Each EO must have at least one EC. EC results do not factor into a trainee's successful completion of the PO.

3. Performance Checks (PCs). PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 Conduct of Instruction Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:

   a. Early warning of trainee difficulties;
   b. Feedback on the effectiveness of training;
   c. Information for the Training Review Board (TRB) (if required); and
   d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall not be given no less than 24 hours of the trainee being informed of their EC or PC failure.

TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 Training Review Board Process.

COURSE REPORTS

10. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51.

11. Trainees will be graded pass or fail. NPTORD 4500-51 Course Reporting and Certificates, provides direction and guidelines for Letter Grading and Ranking.

12. Narratives will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. A record of trainee counselling is to be maintained by the TE.

Task List

Task #	Task Statement	Trg / No Trg	PO
A0005	Perform communications circuit functional checks (comms check)	T	003
A0010	Operate communications equipment	T	003
A0020	Perform Search and Rescue/Distress/SUBSMASH communications procedures	T	003
A0025	Encode/Decode tactical information (with Authenticate communication transmissions)	T	003
A0030	Authenticate communication transmissions	T	003
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	T	002
B0010	Plot formations, screens and dispositions	T	003
B0015	Maintain Stateboards	T	003
B0020	Enforce Emission control (EMCON) restrictions	T	003
B0025	Adhere to Emission control (EMCON) restrictions	T	003
B0030	Contribute to intelligence gathering Ops	T	003
B0035	Display intelligence via UWW brief	T	002
B0040	Disseminate intelligence information	T	002
B0045	Identify Capabilities and limitations of platforms	T	002
B0050	Operate CS	T	003
B0055	Maintain operational logs	N/T	
B0060	Assist with tests and trials with outside agencies	N/T	
C0005	Supervise HMS operations in a mine avoidance environment	N/T	
C0010	Supervise mine detection team in a mine avoidance environment	N/T	
C0025	Operate ancillary acoustic equipment	N/T	
C0040	Analyzing XBT/XSV Forms	N/T	
C0065	Analyze passive acoustic data	N/T	
C0070	Track acoustic contacts	N/T	
C0080	Classify passive acoustic data	N/T	
C0085	Localize passive acoustic contracts	T	003
C0086	Setup Target Motion Analysis Plot	T	003
C0090	Plot Target Motion Analysis (TMA) data	T	003

C0095	Analyze TMA data	T	003
C0100	Construct Contact Area of Probability (AOP)	T	003
C0105	Correlate active/passive/non-acoustic information	T	003
C0115	Calculate acoustic sensor range information	N/T	
C0120	Calculate array tow speeds, tilt and/or balancing	T	002
C0127	Ensure Towed array sonar system is set up	T	003
C0135	Perform noise rounds	N/T	
C0140	Coordinate the employment of UWW weapons and sensors	T	003
C0150	Determine Oceanographic effects on acoustic operations	T	002
C0155	Direct towed array system operations	N/T	
C0160	Establish capabilities and limitations of towed array operations in operation areas (OPAREAS)	T	002
C0195	Perform Passive Aural Listening (PAL)	N/T	
D0045	Prepare UWW exercise support requests	T	002
D0050	Assess UWW threat information	T	003
D0070	Optimize ship's acoustic signature	N/T	
E0005	Compile ASW tactical picture	T	003
E0015	Detect radar contacts	T	003
E0020	Amplify track information	T	003
F0005	Evaluate correlated acoustic information	T	003
F0015	Identify changes in contact's readiness and tactical states	T	003
F0045	Interpret oceanographic data	T	002
G0010	Plan/Design training exercises	T	001
G0040	Compile/Distribute unit OJT/SDP/PIP reports	T	001
G0045	Assist in individual training events	T	001
G0050	Conduct individual training events	T	001
G0055	Correct individual training deficiencies	T	001
G0060	Assist in team training events	T	001
G0065	Conduct team training events	T	001
G0070	Correct team training deficiencies	T	001
G0080	Conduct continuation/proficiency training	T	001
G0085	Conduct unit OJT/SDP/PIP	T	001
G0090	Input training scenarios	N/T	
G0095	Operate simulator/trainer systems	N/T	

G0100	Provide recommendations to PRB/TRB	T	001
G0115	Assess individual training performance	T	001
G0120	Assess team training performance	N/T	
G0140	Provide section/unit with feedback	N/T	
L0005	Conduct Programme evaluation (alignment of docs, availability of trg materials)	N/T	
L0010	Administer course critique process	N/T	
L0015	Track and monitor action items identified during course evaluations	N/T	
L0035	Recommend improvement to training	N/T	

Record of Decision

Dates: 17 November 2020

QSP Amendment

1. IAW CANFORGEN 112/20 RCN JR Rank Designation Change 041200Z SEP 20, all previous RCN JR Rank designations in this QSP have been amended to reflect the new policy.

2. Amendments made to the lesson Enabling Objective naming convention. Changed from Number with Letter to just number. This was to align with NFS Scheduling program requirements.

3. Addition of previously omitted Chapter 3: Assessment of Trainee.

Dates: 4-15 March 20019

Location: S37, rm 305 CFB Stadaconna

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Tallman	J.	CPO2	00324	NTDC(A)
Hamilton	C.	PO2	00324	NTDC(A)
Rudman	B.	PO2	00324	NFS(A)
Hodder	M.	PO2	00324	ADAC
Thompson	R.	MS	00324	PCC(A)
Jawornicki	B.	SLt	00211	NTDC(A)

Working Group Objective:

1. The objective of this working group is to review the RQ-PO2 SONAR OP QSP and align the training with the current Master Task List items as well as reflect the job. It was determined through a Combat Training Review that the current QSP lacked assessments and some structural requirements to direct training.

Specification Task Analysis- No Trains:

Task #	Task Statement	Substantiation
B0055	Maintain operational logs	Instructed at the RQ-S3 level, no further training required.
B0060	Assist with tests and trials with outside agencies	To be instructed on RQ MS coursing, requires RQ PO2's to liaise with outside units.
C0005	Supervise HMS operations in a mine avoidance environment	Instructed on the RQ MS course.
C0010	Supervise mine detection team in a mine avoidance environment	This is a UWWD duty and is instructed on the RQ PO1 / UWWD course.
C0025	Operate ancillary acoustic equipment	Instructed at the RQ-S3 level, no requirement to be re-instructed.

C0040	Analyzing XBT/XSV Forms	Instructed at the RQ-S3 and RQ-S1 levels, no requirement to be re-instructed.
C0065	Analyze passive acoustic data	Instructed at the RQ-S3 to RQ MS levels, no requirement to be re-instructed.
C0070	Track acoustic contacts	Instructed at the RQ-S3 and RQ-S1 levels, no requirement to be re-instructed.
C0080	Classify passive acoustic data	Instructed at the RQ-S3 and RQ-S1 levels, no requirement to be re-instructed.
C0115	Calculate acoustic sensor range information	Instructed at the RQ-S1 level, no requirement to be re-instructed.
C0135	Perform noise rounds	Instructed at the RQ-S3 level, no requirement to be re-instructed.
C0155	Direct towed array system operations	Instructed at the at the RQ MS level, no requirement to be re-instructed.
C0195	Perform Passive Aural Listening (PAL)	Instructed at the RQ-S3 to RQ MS levels, no requirement to be re-instructed.
D0070	Optimize ship's acoustic signature	This is a UWWD duty and is instructed on the RQ PO1 / UWWD course.
G0090	Input training scenarios	To be instructed on RQ MS coursing, no requirement to be re-instructed.
G0095	Operate simulator/trainer systems	To be instructed on RQ MS coursing, no requirement to be re-instructed.
G0120	Assess team training performance	This is a task that should be at the RQ PO1 / UWWD level.
G0140	Provide section/unit with feedback	No requirement for instruction on how to perform this task.
L0005	Conduct Programme evaluation (alignment of docs, availability of trg materials)	Task is specific to training organizations.
L0010	Administer course critique process	Task is specific to training organizations.
L0015	Track and monitor action items identified during course evaluations	Task is specific to training organizations.
L0035	Recommend improvement to training	Task is specific to training organizations.

Discussions / Decisions:

2. An examination of the T/NT list that was developed in the Combat Training Review resulted in a general agreement of the T/NT decisions.
    a. The group determined that F0045 Interpret oceanographic data needed to be included as a train at the PO2 level based on the increased use of oceanographic knowledge regarding making recommendations at a tactical level. The employment of sensors and air assets is based on the advanced knowledge of the oceanographic picture.
    b. G0100 Provide recommendation to PRB/TRB was changed to a train based on the training PO position on the ship who may have a student conducting an OJPR with insufficient progress. The duty area of being a training PO determined this as a train.
3. Group discussed the requirement of drafting messages at the PO2 level. It was determined by the board that basic unit level message preparations are conducted at the PO2 level and should be taught (to include RSS, RFI, and OCS).

4. When identifying the requirement to review CASEX relaxation tables, there was discussion regarding the use of a program to assist in the creation of such tables? The board determined that the program not been updated and is not considered a mandatory requirement WRT the publication and therefore should not be taught on the course.
5. Throughout the creation of the various lessons, much consideration was put into the bicoastal model and ensuring all tools and facilities were available to both. Any tours or recourses that are not mirrored, there was a specific lecture identified as a backup to ensure the teaching points were covered. There was an attempt to mitigate any coastal discrepancies however it was noted that if the tours were accessible it would be of benefit, therefore leaving it out would be a detriment to those who could benefit from the tour facilities.

Course History:

6. In 2005: a course duration amendment was conducted to lengthen the trainer time allocated to a new capability (NIXIE) to 4 days as the Main Control Unit and Remote Control Unit operations needed to be covered and provided the necessary practical time to become proficient.
7. The P-2 amendment 10 Oct 06: restructured/regrouped many of the oceanography teaching points based on new tools available to assist in the determination of environmental effects on UWW operations.
8. August 2009: An Amendment was produced for Performance Checklists PO 002, 003A and 003B (CASEX, TDT and MEGA phase). The new Check off lists has incorporated the new TDT procedures as the required standard. The former checklist was vague at times allowing individual assessors a wide interpretation of how many marks to give the trainee for a completed task. All three Check lists now specifically allocates how many marks the assessor grants to each trainee for a completed task (marking scheme broken down more).
9. May 2011: An Amendment was produced for performance Checklists PO 002, 003A and 003B (CASEX, TDT, and MEGA phase). Slight modifications were made to the CASEX and TDT checklist while major amendments were added to the MEGA phase. The Mega Phase will now be based upon a “SAT” or “UNSAT” instead of the numerical weight value in the past. A Pass mark of SAT’ will be based upon the Final Assessment Board. The course has also been reduced by a total of 5.4 training days. This was due to the removal of the AN-SLQ 25, HITASS , TAT training systems and students being on their respective coasts (East and West) for instruction of PO 001.
10. May 2013: HCM Hybrid amendment - changes were designed to make this QSP HCM Hybrid Compliant and ensure capabilities and limitations of new HCM driven equipment are covered. Changes was the inclusion of:

    a. Capabilities and limitations of CMS330 into EO 002.1
    b. SHINCOM 3100 into EO 002.1

11. In 2019, NTDC(A) was ordered to conduct a Combat Training Review, IOT identify training deficiencies within the QSPs currently used for instruction. The RQ-PO2 course had an interim QSP that had been drafted and approved at the NTDC(A) level, with no further approval from NPTG (HQ) as the authority had been delegated during the Campus transition. The 2019 version of the QSP was drafted using a QSP working group including SMEs and a TDO to create a new document based on including a Training Petty Officer component and address the message drafting components that were necessary. With the removal of the MEGA, many courses needed to reconsider their training structure and timings to focus solely on the RQ coursing requirements.
12. October 2020: Amendment to RCN rank changes IAW CANFORGEN 112/20 RCN 023/20 041200Z SEP 20.

Costing Information

Based on 12 students total, 6 West Coast and 87 training days (plus weekends 121 calendar days)

Course: RQ-PO2 SONAR	Serial:					
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	6	$1,600.00		$9,600	$9,600	$96,000.00
R&Q	6	$78.75	121	$57,173	$57,173	$571,725.00
Incidentals	6	$17.30	121	$12,560	$12,560	$125,598.00
Subtotal				**$79,332.30**	**$79,332.30**	**$793,323.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	12	$50.00	1	$600	$600	$6,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	12	$0.00	1	$0	$0	$0.00
IT Equip (Proj, Smart board etc)	3	$0.00	1	$0	$0	$0.00
Instructional Materials (Printed / Handouts etc.)	14	$30.00	1	$420	$240	$2,580.00
Subtotal				**$1,020**	**$840**	**$8,580.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1) (incremental)5	0	$0.00	0	$0	$0	$0.00
Instructor (Lt(N)) (incremental)5	0	$0.00	0	$0	$0	$0.00
Incremental Civilian Instructor	0	0	0	$0	$0	$0.00
		subtotal		**$0.00**	**$0.00**	**$0.00**
	Total Estimated Costs			**$80,352**	**$80,172**	**$801,903**

Assessment Plan

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess student's competence in planning training exercises	Practical -creation of a training calendar	Student must obtain 9/12 SATs IAW EC 001.01 Checklist
EC 001.02	EO 001.02	To assess student's competence in conducting OJT, SDP and PIP training	Practical -(create Monitor Mass Training Tracker and Jeopardy alert)	Student must obtain 6/8 SATs IAW EC 001.02 Checklist
PC 001	PO 001	To assess student's competence in performing the duties of a training Petty Officer	Practical -(creation of a training calendar and feedback notes)	Student must obtain 12/16 SATs IAW PC 001 Checklist
EC 002.01	EO 002.01	To assess student's competence in preparing UWW messages	Practical – creation of a TABORD and RSS/UWON	Student must obtain 18/25 SATs IAW EC 002.01 Checklist
EC 002.02	EO 002.02	To assess student's competence in displaying intelligence via UWW brief	Practical- create intel/oceanographic brief	Student must obtain 14/19 SATs IAW EC 002.02 Checklist
PC 002	PO 002	To assess student's competence in assisting in mission/exercise preparations	Practical – submit TABORD and intel/oceanographic brief	Student must obtain 25/33 SATs IAW PC 001 Checklist
EC 003.01	EO 003.01	To assess student's competence in operating communications equipment	Practical Voice Procedures	Student must obtain 15/20 SATs IAW EC 003.01 Checklist
EC 003.02	EO 003.02	To assess student's competence in operating Command System	Practical	Student must obtain 40/50 SATs IAW EC 003.02 Checklist
EC 003.03.01	EO 003.03	To assess student's competence in coordinating the employment of UWW weapons, sensors and decoys (CASEX)	Practical	Student must obtain 26/34 SATs IAW EC 003.03A Checklist
EC 003.03.02	EO 003.03	To assess student's competence in coordinating the employment of UWW weapons, sensors and decoys (NIXIE)	Practical	Student must obtain 10/13 SATs on non-critical factors AND 1/1 on critical factors IAW EC 003.03B Checklist
EC 003.03.03	EO 003.03	To assess student's competence in coordinating the employment of UWW weapons, sensors and decoys (TDT)	Practical	Student must obtain 10/13 SATs on non-critical factors and 2/2 on critical factors IAW EC 003.03C Checklist
PC 003	PC 003	To assess student's competence in operating communications equipment	Practical	Student must obtain 27/36 SATs for section 1-3 and 9/11 SATs on non-critical factors and 2/2 on critical factors for section 4 IAW PC 003 Checklist

TEL Development Opportunities

Title	Type	Description of Requirement	Lesson	SOR #
PC 001 bathy video	video	Video of Junior operator launching a bathy	PC 001	

Scheduling Information

Scheduling Annex - SONAR OP RQ-PO2													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL / Critical Resources	Sec Class	WB	Desks	Type	Qual	Sec Cl
PO 001 - Perform the Duties of a Training Petty Officer													
EO 001.01.01	50	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	NTDC or TDO	Sec Cl
EO 001.01.02	50	Classroom	13	0	0	1	DWAN, SONAR OP RQ-OS/AB QSP	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 001.01.03	50	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 001.01.05	50	Classroom	13	0	0	1	Guest Lecture - CCFL/P or Sea Training	UNC	1	13	GS	CCFL/P or STG SONAR OP	Sec Cl
EO 001.01.05	100	Classroom	13	0	0	1	Whiteboards / Flip charts	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 001.01.06	200	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 001.01.07	50	Classroom	13	0	0	1	Whiteboard, DWAN	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 001.01.08	100	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EC 001.01	100	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	RQ SONAR OP PO2	Sec Cl
EO 001.02.01	50	Classroom	13	0	0	1	DWAN & Monitor Mass access	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 001.02.02	150	Classroom	13	0	0	1	DWAN with Projector	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 001.02.03	50	Classroom	13	0	0	1	DWAN, Guest Lecture ADAC	UNC	1	13	GS	ADAC SONAR OP RQ PO2	Sec Cl
EO 001.02.04	50	Classroom	13	0	0	1	DWAN, Guest Lecture TDO	UNC	1	13	Mil	TDO	Sec Cl
EO 001.02.05	100	Classroom	13	0	0	1	DWAN, Flip charts or Whiteboard	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EC 001.02	50	Classroom	13	0	0	1	DWAN & Monitor Mass access	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
PC 001	100	Classroom	13	0	0	1	DWAN with Projector & Speakers	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
PO 002 - Assist in Mission/Exercise Preparations													
EO 002.01.01	150	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	CONF	1	13	Mil	UWWD	Sec Cl
EO 002.01.02	100	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.03	250	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	CONF	1	13	Mil	UWWD	Sec Cl
EO 002.01.04	100	Classroom	0	13	0	1	CSNI access	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 002.01.05	500	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.06	250	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EC 002.01	150	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.02.01	350	TRINITY / Classroom	6	6	0	1	Tour of TRINITY or prepared Guest lecture	SEC	1	13	GS	Any TRINITY staff	Sec Cl
EO 002.02.02	150	Classroom	6	6	0	0	CSNI access	SEC	0	0	Mil	SONAR OP RQ PO2	Sec Cl
EO 002.02.03	100	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 002.02.04	250	Classroom	6	6	0	1	DWAN / CSNI	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 002.02.05	300	Classroom	0	13	0	1	CSNI with projector	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EC 002.02	100	Classroom	13	13	13	1	DWAN & CSNI or Secret stand alone	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
PC 002	200	Classroom	13	13	13	1	DWAN & CSNI or Secret stand alone	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl

PO 003 - Contribute to Intelligence gathering Operations													
EO 003.01.01	50	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	GS	NCI OP RQ MS	Sec Cl
EO 003.01.02	100	SEACOT	0	0	0	0	SEACOT with SHINCOM terminals	SEC	0	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.01.03	100	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	GS	NCI OP RQ MS	Sec Cl
EO 003.01.04	250	SEACOT	0	0	0	0	SEACOT with SHINCOM terminals	SEC	0	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.01.05	100	Classroom	13	0	0	1	DWAN with projector	UNC	0	13	GS	CANSUBFOR Officer	Sec Cl
EC 003.01	300	SEACOT	0	0	0	0	SEACOT with SHINCOM terminals	SEC	0	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.03.01	200	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.02	150	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.03	100	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.04	50	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.05	100	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	GS	SAC or UWWD	Sec Cl
EO 003.03.06	100	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.07	50	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.08	50	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.09	50	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.03.10	100	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	GS	IUSS or UWWD & Post ORO	Sec Cl
EO 003.02.01	50	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.02.02	900	SEACOT	0	0	0	1	SEACOT with Instructor terminal	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.02.03	3600	SEACOT	0	0	0	1	SEACOT with Instructor terminal	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EC 003.02	150	SEACOT	0	0	0	1	SEACOT	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.03.11	4200	Classroom	0	0	0	0	SEACOT with ASW Team	SEC	1	0	Mil	SONAR OP RQ PO2	Sec Cl
EC 003.03.01	1200	Classroom	0	0	0	0	SEACOT with ASW Team	SEC	0	0	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.03.12.1	100	Classroom	0	13	0	1	CSNI classroom	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.03.12.2	300	NIXIE trainer	0	0	0	0	NIXIE trainer	SEC	0	0	Mil	SONAR OP RQ PO2	Sec Cl
EC 003.03.02	200	NIXIE trainer	0	0	0	0	NIXIE trainer	SEC	0	0	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.03.13	300	Classroom	0	13	13	1	CSNI or Secret stand alone workstations	SEC	1	13	Mil	SONAR OP RQ PO2	Sec Cl
EO 003.03.14	3000	SEACOT	0	0	0	0	SEACOT with ASW Team	SEC	1	0	Mil	SONAR OP RQ PO2	Sec Cl
EC 003.03.03	650	SEACOT	0	0	0	0	SEACOT with ASW Team	SEC	1	0	Mil	SONAR OP RQ PO2	Sec Cl
PO 003	6000	SEACOT	0	0	0	0	SEACOT with ASW Team	SEC	1	0	Mil	SONAR OP RQ PO2	Sec Cl
PC 003	3600	SEACOT	0	0	0	0	SEACOT with ASW Team	SEC	1	0	Mil	SONAR OP RQ PO2	Sec Cl

Totals	30350
Periods	607
Days	75.875

Plan, Train, Do

PO	EO	Maintain Readiness	Plan	Train	Execute
001	01	Plan/Design Training Exercises		Plan/Design Training Exercises	
	02	Conduct unit OJT, SDP, Pip and continuation/proficiency training		Conduct unit OJT, SDP, Pip and continuation/proficiency training	
002	01		Prepare UWW messages		
	02		Display Intelligence via UWW brief		
003	01				Operate Communications Equipment
	02				Operate Command System
	03				Coordinate the Employment of UWW weapons, sensors and decoys

Recommended Schedule

Time	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Period 1 0800-0850	**Expectations/Admin ISSO, IT, Command Intro (BAT)**	**Section resources – Equipment & expendables** EO 001.01.05	**Developing a Training Calendar** EO 001.01.08	**EC 001.01 Plan/Design Training Exercises**	**EC 001.02 EC Conducting unit OJT, SDP, PIP and continuation/ proficiency training**
Period 2 0850-0940	**Pre Course Admin**, BAT/TAT				**Interpreting OPTASK ASW/SCC** 002.01.02
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Mission Intro** Provide Mission specific message traffic and famil with course mission	**Section resources - personnel** EO 001.01.06	**Utilize Monitor Mass Training Tracker** EO 001.02.01	**Team Training** EO 001.02.05	**Interpreting OPTASK ASW/SCC** 002.01.02
Period 4 1040-1130	**Occupational Progression** EO 001.01.01 Guest Lecture: TDO/TCD		**OPJR Evaluation Standards** EO 001.02.02		**Interpreting TABORDs and CASEX Relaxation Tables** 002.01.03
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**RQ-S2 OJPR** EO 001.01.02	**Section resources - personnel** EO 001.01.06	**OPJR Evaluation Standards** EO 001.02.02	**Interpreting OPGEN & OPSTAT UNIT** 002.01.01	**Interpreting TABORDs and CASEX Relaxation Tables** 002.01.03
Period 6 1320-1410	**CRRs** EO 001.01.03				
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Ship/Fleet Training** EO 001.01.04 Guest Lecture: CCFL/Sea Trg	**Scheduling Training Events with outside units** EO 001.01.07	**Monthly Acoustic Analysis Training Packs** EO 001.02.03	**Interpreting OPGEN & OPSTAT UNIT** 002.01.01	**Interpreting TABORDs and CASEX Relaxation Tables** 002.01.03
Period 8 1510-1600	**PT**	**PT**	**PRB/TRB process** EO 001.02.04 Guest Lecture: TDO	**PT**	

Time	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**PC 001** **Perform the Duties of a Training Petty Officer**	**Drafting water space management messages** EO 002.01.05	**Drafting TBORDs** EO 002.01.05	**BAT/TAT**	**EC 001.02 EC Title**
Period 2 0850-0940				**PT**	**Identify Caps/Lims of Platforms** 002.02.02
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Drafting RFIs** EO 002.01.04	**Drafting water space management messages** EO 002.01.05	**Drafting TABORDs** EO 002.01.05	**EC 002.02** **Prepare UWW messages**	**Identify Caps/Lims of Platforms** 002.02.02
Period 4 1040-1130			**PT**		
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Drafting water space management messages** EO 002.01.05	**Drafting water space management messages** EO 002.01.05	**Interpreting Intelligence Data** EO 002.02.01 Trinity/ADAC Tour	**EC 002.02** **Prepare UWW messages**	**Interpret Oceanographic Data** 002.02.03
Period 6 1320-1410				**Interpreting Intelligence Data** EO 002.02.01	
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Drafting water space management messages** EO 002.01.05	**Drafting TBORDs** EO 002.01.06	**Interpreting Intelligence Data** EO 002.02.01 Trinity/ADAC Tour	**Interpreting Intelligence Data** EO 002.02.01	**BAT/TAT**
Period 8 1510-1600					**PT**

ANNEX J – RECOMMENDED SCHEDULE

Time	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**Determining Oceanographic Effect** EO 002.02.04	**Building Intel/Oceanographic Brief** EO 002.02.05	**EC 002.02** **Display Intelligence via UWW brief**	**PC 002** **Assist in Mission/Exercise Preparations**	**EC 003.01** **Operate Communications Equipment**
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Determining Oceanographic Effect** EO 002.02.04	**Building Intel/Oceanographic Brief** EO 002.02.05	**Tactical Communications** EO 003.01.03	**PC 002** **Assist in Mission/Exercise Preparations**	**Compile ASW Picture** 003.03.01 Guest Lecture: UWWD
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Determining Oceanographic Effect** EO 002.02.04	**BAT/TAT**	**Voice Reports** EO 003.01.04	**Voice Reports** EO 003.01.04	**Compile ASW Picture** 003.03.01 Guest Lecture: UWWD
Period 6 1320-1410	**Building Intel/Oceanographic Brief** EO 002.02.05	**Basic Communications Theory** EO 003.01.01 Guest Lecture: MS NCIOP			
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Building Intel/Oceanographic Brief** EO 002.02.05	**Operate Internal/External SHINCOM Nets** EO 003.01.02	**Voice Reports** EO 003.01.04	**SUBSAR/Distress/SUB SMASH** 003.01.05 Guest Lecture: CANSUBFOR	**BAT/TAT**
Period 8 1510-1600	**PT**		**PT**		**PT**

Time	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**Tactical Employment of ASW sensors** EO 003.03.02 Guest Lecture: UWWD	**Air Operations** EO 003.03.05 Guest Lecture: SAC/UWWD	**IUSS & Theatre ASW** EO 003.03.10 Guest Lecture: IUSS member/UWWD & Post ORO NWO	**Set Up Command System** EO 003.02.02	**Set Up Command System** EO 003.02.02
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Tactical Employment of ASW sensors** EO 003.03.02 Guest Lecture: UWWD	**ASW Search and Attack Plans** EO 003.03.06 Guest Lecture: UWWD	**Tactical Communications** EO 003.01.03	**Set Up Command System** EO 003.02.02	**Set Up Command System** EO 003.02.02
Period 4 1040-1130	**BAT/TAT**				
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Command Classification** EO 003.03.03 Guest Lecture: UWWD	**ASW Pre-Plans** EO 003.03.07 Guest Lecture: UWWD	**EMCON Enforcement** EO 003.02.01	**Set Up Command System** EO 003.02.02	**Set Up Command System** EO 003.02.02
Period 6 1320-1410		**Surface and A/C attacks** EO 003.03.08 Guest Lecture: UWWD	**Set Up Command System** EO 003.02.02		
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Water space management** EO 003.03.04 Guest Lecture: UWWD	**Anti-Ship Torpedo Defence** EO 003.03.09 Guest Lecture: UWWD	**Set Up Command System** EO 003.02.02	**Set Up Command System** EO 003.02.02	**Set Up Command System** EO 003.02.02
Period 8 1510-1600	**PT**	**BAT/TAT**	**PT**		

Time	DAY 21	DAY 22	DAY 23	DAY 24	DAY 25
Period 1 0800-0850	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Setting Up Tactical Decisions Aids** EO 003.02.03	**PT**	**Setting Up Tactical Decisions Aids** EO 003.02.03	**BAT/TAT**	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 8 1510-1600	**PT**	**BAT/TAT**		**PT**	

Time	DAY 26	DAY 27	DAY 28	DAY 29	DAY 30
Period 1 0800-0850	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Setting Up Tactical Decisions Aids** EO 003.02.03	**PT**	**Setting Up Tactical Decisions Aids** EO 003.02.03	**BAT/TAT**	**Setting Up Tactical Decisions Aids** EO 003.02.03
Period 8 1510-1600	**PT**	**BAT/TAT**		**PT**	

ANNEX J – RECOMMENDED SCHEDULE

Time	DAY 31	DAY 32	DAY 33	DAY 34	DAY 35
Period 1 0800-0850	**Setting Up Tactical Decisions Aids** EO 003.02.03	**EC 003.02 Operate Command System**	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Perform Duties of SCS during CASEX** EO 003.03.11	**EC 003.02 Operate Command System**	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 4 1040-1130		**Perform Duties of SCS during CASEX** EO 003.03.11			
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Perform Duties of SCS during CASEX** EO 003.03.11	**PT**	**Perform Duties of SCS during CASEX** EO 003.03.11	**BAT/TAT**	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 8 1510-1600	**PT**	**BAT/TAT**		**PT**	

Time	DAY 36	DAY 37	DAY 38	DAY 39	DAY 40
Period 1 0800-0850	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Perform Duties of SCS during CASEX** EO 003.03.11	**PT**	**Perform Duties of SCS during CASEX** EO 003.03.11	**BAT/TAT**	**Perform Duties of SCS during CASEX** EO 003.03.11
Period 8 1510-1600	**PT**	**BAT/TAT**		**PT**	

ANNEX J – RECOMMENDED SCHEDULE

Time	DAY 41	DAY 42	DAY 43	DAY 44	DAY 45
Period 1 0800-0850	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**Perform Duties of SCS during CASEX** EO 003.03.11	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Perform Duties of SCS during CASEX** EO 003.03.11	**PT**	**BAT/TAT**	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**
Period 8 1510-1600	**PT**	**BAT/TAT**	**PT**		

Time	DAY 46	DAY 47	DAY 48	DAY 49	DAY 50
Period 1 0800-0850	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**NIXIE** EO 003.03.12	**EC 003.03.02 Coordinate the Employment of UWW weapons, sensors and decoys (NIXIE)**	**Torpedo Detect and Track** EO 003.03.13	**Torpedo Detect and Track Practical** EO 003.03.14
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**NIXIE** EO 003.03.12	**EC 003.03.02 Coordinate the Employment of UWW weapons, sensors and decoys (NIXIE)**	**Torpedo Detect and Track** EO 003.03.13	**Torpedo Detect and Track Practical** EO 003.03.14
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**NIXIE** EO 003.03.12	**Torpedo Detect and Track** EO 003.03.13	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EC 003.03.01 Coordinate the Employment of UWW weapons, sensors and decoys (CASEX)**	**NIXIE** EO 003.03.12	**BAT/TAT**	**BAT/TAT**	**Torpedo Detect and Track Practical** EO 003.03.14
Period 8 1510-1600			**PT**	**PT**	**PT**

Time	DAY 51	DAY 52	DAY 53	DAY 54	DAY 55
Period 1 0800-0850	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Torpedo Detect and Track Practical** EO 003.03.14	**PT**	**BAT/TAT**	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14
Period 8 1510-1600	**PT**	**BAT/TAT**	**PT**		

Time	DAY 56	DAY 57	DAY 58	DAY 59	DAY 60
Period 1 0800-0850	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**EC 003.03.03 Coordinate the employment of UWW weapons, sensors and decoys (TDT)**	**EC 003.03.03 Coordinate the employment of UWW weapons, sensors and decoys (TDT)**
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**EC 003.03.03 Coordinate the employment of UWW weapons, sensors and decoys (TDT)**	**EC 003.03.03 Coordinate the employment of UWW weapons, sensors and decoys (TDT)**	**EC 003.03.03 Coordinate the employment of UWW weapons, sensors and decoys (TDT)**
Period 4 1040-1130					**Consolidation: Perform the duties of SCS during Operations** 003
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**EC 003.03.03 Coordinate the employment of UWW weapons, sensors and decoys (TDT)**	**EC 003.03.03 Coordinate the employment of UWW weapons, sensors and decoys (TDT)**	**Consolidation: Perform the duties of SCS during Operations** 003
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Torpedo Detect and Track Practical** EO 003.03.14	**Torpedo Detect and Track Practical** EO 003.03.14	**BAT/TAT**	**BAT/TAT**	**Consolidation: Perform the duties of SCS during Operations** 003
Period 8 1510-1600	**PT**		**PT**	**PT**	

ANNEX J – RECOMMENDED SCHEDULE

Time	DAY 61	DAY 62	DAY 63	DAY 64	DAY 65
Period 1 0800-0850	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**BAT/TAT**	**BAT/TAT**	**Consolidation: Perform the duties of SCS during Operations** 003
Period 8 1510-1600	**PT**		**PT**	**PT**	

Time	DAY 66	DAY 67	DAY 68	DAY 69	DAY 70
Period 1 0800-0850	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**BAT/TAT**	**BAT/TAT**	**Consolidation: Perform the duties of SCS during Operations** 003
Period 8 1510-1600	**PT**		**PT**	**PT**	

Time	DAY 71	DAY 72	DAY 73	DAY 74	DAY 75
Period 1 0800-0850	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**BAT/TAT**	**BAT/TAT**	**Consolidation: Perform the duties of SCS during Operations** 003
Period 8 1510-1600	**PT**		**PT**	**PT**	

ANNEX J – RECOMMENDED SCHEDULE

Time	DAY 76	DAY 77	DAY 78	DAY 79	DAY 80
Period 1 0800-0850	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Consolidation: Perform the duties of SCS during Operations** 003	**Consolidation: Perform the duties of SCS during Operations** 003	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**Consolidation: Perform the duties of SCS during Operations** 003	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**BAT/TAT**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**BAT/TAT**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 8 1510-1600	**PT**		**PT**	**PT**	

ANNEX J – RECOMMENDED SCHEDULE

Time	DAY 81	DAY 82	DAY 83	DAY 84	DAY 85
Period 1 0800-0850	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 2 0850-0940					
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 4 1040-1130					
1130-1230	LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 6 1320-1410					
1410-1420	BREAK	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**BAT/TAT**	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**	**BAT/TAT**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 8 1510-1600	**PT**		**PT**	**PT**	

Time	DAY 86	DAY 87
Period 1 0800-0850	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 2 0850-0940		
0940-0950	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 Contribute to Intelligence Gathering Operations**
Period 4 1040-1130		
1130-1230	LUNCH	**LUNCH**
Period 5 1230-1320	**PC 003 Contribute to Intelligence Gathering Operations**	**PC 003 RETEST**
Period 6 1320-1410		
1410-1420	BREAK	**BREAK**
Period 7 1420-1510	**PC 003 Contribute to Intelligence Gathering Operations**	**BAT/TAT Post Course Dialogue**
Period 8 1510-1600		